The Planetary Science Journal, 5:245 (36pp), 2024 November https://doi.org/10.3847/PSJ/ad8130

Analysis of Enceladus's Time-variable Space Environment to Magnetically Sound its Interior

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Received 2024 May 24; revised 2024 September 25; accepted 2024 September 26; published 2024 November 14

Abstract

We provide a comprehensive study of Enceladus's time-variable magnetic field environment and identify in measurements of the Cassini spacecraft signatures that appear to be consistent with induced fields from the moon's interior. Therefore, we first analyze the background field Enceladus is exposed to within 21 flybys and 50 crossings of the moon's orbit by the Cassini spacecraft. Considering magnetic field variability due to Enceladus's eccentric orbit, Saturn's planetary period oscillations, and local time effects within the magnetospheric current sheet, we construct predictive, time-variable background fields near Enceladus with a correlation coefficient of 0.75 and larger compared to the measured background fields. Subsequently, we build a geophysically based electrical conductivity model of Enceladus's ocean from the equation of state for saline water and mixing laws for a porous core permeated by water. Using this conductivity model and the derived time-variable fields, we calculate expected induced fields. For close flybys, we identify within mostly plume-dominated magnetic field perturbations of 10–30 nT much smaller perturbations of 1–3 nT, which could be consistent with induction. The flybys over Enceladus's north pole are best suited for induction studies, and the associated measurements suggest that a conductivity of the ocean with 1–3 S m^{-1} is not sufficient to produce an adequate induction response, but they support a highly conductive, porous core of 20–30 S m^{-1} and/or a more conductive ocean. Our study also provides strategies for future magnetic sounding of Enceladus.

Unified Astronomy Thesaurus concepts: Enceladus (2280); Magnetic fields (994)

1. Introduction

Magnetic induction studies are a powerful means to explore the interior of planetary bodies and to search for subsurface oceans within icy moons. This technique has been very successfully applied at the icy moons of Jupiter, where the dominant inducing effect stems from the tilt of Jupiter's dipole magnetic field (K. K. Khurana et al. 1998; F. M. Neubauer 1998; C. Zimmer et al. 2000). The azimuthal symmetry of Saturn's internal magnetic field (M. K. Dougherty et al. 2018), however, does not provide a similar inducing, time-variable magnetic field to that in the Jupiter system. The first part of our study therefore analyzes the magnetic field and plasma environment Enceladus is exposed to in order to find other means of time-variable fields for electromagnetic sounding of its interior.

Enceladus is a solar system object that possesses all key properties to make it a habitable planetary body (e.g., reviews by M. L. Cable et al. 2021; G. Choblet et al. 2022; A. H. Sulaiman et al. 2022 or F. Postberg et al. 2023). All six elements considered essential for habitability, i.e., carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur, have been demonstrated to be present in Enceladus's subsurface ocean (F. Postberg et al. 2018, 2023). Theses conditions and the fact that the subsurface ocean directly sources its plumes make Enceladus an ideal target for the exploration of habitable worlds other than Earth. Essential progress in understanding Enceladus came with the detection of the plume activity stemming from four major rifts within the ice crust near Enceladus's south pole, which is one of the great discoveries of the Cassini mission (M. K. Dougherty et al. 2006; C. C. Porco et al. 2006; J. R. Spencer et al. 2006; R. L. Tokar et al. 2006; J. H. Waite et al. 2006).

The plume activity is time variable (J. Saur et al. 2008) and was demonstrated to be quasi-periodic with maximum activity when Enceladus is near its apocenter on its slightly eccentric orbit (M. M. Hedman et al. 2013). Enceladus's plumes are sourced by a saline subsurface water reservoir based on plume grains that contain 0.5%–2% sodium and potassium salts by mass (F. Postberg et al. 2009, 2011). Gravity measurements of Enceladus demonstrated the presence of an at least regional subsurface sea extending to 50° south latitude (L. Iess et al. 2014). The global extent of the ocean was later inferred from observation and analysis of the Enceladus's physical libration (P. C. Thomas et al. 2016).

Several authors derived models of Enceladus's interior based on measurements of gravity, libration, and physical shape (e.g., P. C. Thomas et al. 2016; T. Van Hoolst et al. 2016; R. Tajeddine et al. 2017; R. S. Park et al. 2024). The overall consensus from theses models is that Enceladus, with a radius of 256.6 km, consists of a rocky core with radius 180–200 km and mass density in the range 2300–2500 kg m^{-3} and an outer water layer with thickness 50–70 km, separated into an ice shell and a subsurface ocean (R. S. Park et al. 2024). An average ice shell thickness of 21 km and an ocean thickness of 37 km have been predicted by P. C. Thomas et al. (2016), while the model of R. S. Park et al. (2024) finds a thicker mean ice shell of 27–33 km and a thinner mean ocean thickness of 21–26 km. The ocean is estimated to be maintained by more than 10 GW of heat generated by tidal friction inside an unconsolidated, porous rocky core (G. Choblet et al. 2017). Due to upwelling of hot water from the core, the icy shell is expected to be much thinner at Enceladus's poles. Near the south pole the ice crust thickness is expected to be 5 km or less, while near the north pole values of around 10 km (and locally less) are discussed in the literature (e.g., G. Choblet et al. 2017; M. L. Cable et al. 2021).

Despite the measurements of sodium within the ice grains in Enceladus's plumes (F. Postberg et al. 2011), the salinity of Enceladus's ocean is not well constrained, but it has major implications for the ocean circulation (W. Kang et al. 2022a, 2022b; Y. Zeng & M. F. Jansen 2024). For example, the meridional overturning circulation caused by heat and salt exchange with a poleward-thinning ice shell has opposite directions at very low and very high salinities, and thus the salinity also controls the thickness variations of the ice shell (W. Kang et al. 2022b).

The aim of this work is to search for signs of induction from within Enceladus and possibly characterize the salinity of fluids in the interior of the moon. To the authors´ knowledge, this work is the first study applying this technique to Enceladus. Analyzing induction at Enceladus is plagued by two problems. The first is the absence of Jupiter-like predictable magnetic field variations referred to as primary or inducing fields, due to the azimuthal symmetry of Saturn's intrinsic magnetic field. The second problem is the interaction of Saturn's magnetospheric plasma with the gas and dust in Enceladus's plumes. We will show that this plasma interaction generates magnetic field perturbations that are about a factor of 10 or more larger than those expected from induction, the latter fields are referred to as secondary or induced magnetic fields. The plasma magnetic fields have been studied in detail by a number of authors (M. K. Dougherty et al. 2006; K. K. Khurana et al. 2007; J. Saur et al. 2007, 2008; H. Kriegel et al. 2009, 2011; S. Simon et al. 2011). Both problems are key topics to be addressed in this study.

In this work we provide a comprehensive analysis of inducing and induced fields at Enceladus. In the next section we begin with a discussion of possible sources of time-variable fields Enceladus is exposed to (Section 2) and then analyze measurements by the Cassini spacecraft at radial distances of Enceladus's orbit to search for and characterize the predictable and time-periodic magnetic field components (Section 3). Based on the knowledge of Enceladus's interior from the literature, we subsequently develop a three-layer conductivity model with an insulating crust and an ocean and a core with conductivities possibly ranging from 1×10^{-2} to 1×10^{2} S m^{-1} for the latter two. With this model we calculate possible induction responses for the synodic rotation period of Saturn and the orbital period of Enceladus (Section 4). We then search for signatures of induction in the magnetic field measurements obtained during close flybys of the Cassini spacecraft (Section 5). We find that the north polar flybys are best suited to quantitatively analyze the possible induction signals. In this step we use geophysical constraints based on the equation of state of seawater for Enceladus's ocean and a two-phase model for its core to narrow down and further characterize the conductivities of the ocean and the core (Section 7). We end with a discussion of our results and suggestions of strategies for induction sounding of Enceladus in future measurements (Section 8).

2. Sources of Time-periodic Fields at Enceladus

No azimuthal asymmetry of Saturn's internal magnetic field has been observed (M. K. Dougherty et al. 2018). In the case of Jupiter, such asymmetry (often approximated as a tilted dipole) serves as the primary inducing field employed in induction studies of its moons (K. K. Khurana et al. 1998; F. M. Neubauer 1998, 1999; C. Zimmer et al. 2000; J. Saur et al. 2010; M. Seufert et al. 2011). However, magnetospheric processes and Enceladus's orbital properties still cause periodic time-variable fields in the moon's rest frame. In this section, these time-variable fields are discussed (see also Figure 1), before we search for them within magnetometer measurements of the Cassini spacecraft (Section 3).

2.1. Orbital Geometry of Enceladus

Enceladus's orbital eccentricity is $e = 0.005$, and its inclination is $i = 0.0°$ (R. Jacobson 2022). The effect of the small eccentricity on the magnetic field variability at Enceladus can be approximated by Taylor-expanding the dipole component of Saturn's field. This expansion shows that the latitudinal magnetic field varies by $\delta B_\Theta = \pm 3eB_0$, with B_0 Saturn's average latitudinal magnetic field at Enceladus's orbit (see Appendix A). Approximating the field B_0 with 330 nT (M. K. Dougherty et al. 2006), we obtain $\delta B_\Theta = \pm 5.0$ nT. A recent independent analysis by M. J. Styczinski & C. J. Cochrane (2024) found similar values for δB_Θ. The variability of δB_Θ is the primary variability Enceladus is exposed to based on its eccentric orbit. Due to Enceladus's eccentricity, the radial component of the internal field also experiences a small but, for this study, negligible variability of $\delta B_r = \pm 0.2$ nT from the degree $n > 1$ components (M. K. Dougherty et al. 2018). The azimuthal component B_ϕ of Saturn's internal field is zero owing to its azimuthal symmetry. A detailed derivation of the variability of the three magnetic field components is provided in Appendix A. The associated period of the magnetic field variability is not Enceladus's sidereal orbital period $T_{\text{orbit}} = 1.370218$ days (R. Jacobson 2022) but slightly modified owing to the precession of the periapsis with a period of $T_\omega = 2.9$ yr (see also M. J. Styczinski & C. J. Cochrane 2024). The resultant synodic orbital period is, however, only 0.1% larger. In the following, when we refer to the orbital period variability, this effect is included in our calculations of the inducing phases.

2.2. Planetary Period Oscillation Fields

Saturn's magnetospheric field exhibits planetary period oscillations (PPOs) first detected in radio data by the Voyager spacecraft (M. L. Kaiser et al. 1980). The origin of these oscillations is still not fully understood, but they might be caused by atmospheric/ionospheric effects (C. G. A. Smith 2006; X. Jia et al. 2012). The PPOs have been studied extensively and are seen in basically all observed magnetospheric quantities, such as the magnetic field (G. Giampieri et al. 2006; D. J. Andrews et al. 2008; G. Provan et al. 2009), Saturn's kilometric radiation (P. H. M. Galopeau & A. Lecacheux 2000; L. Lamy et al. 2008), plasma density (D. A. Gurnett et al. 2007), and even turbulence properties (M. von Papen & J. Saur 2016).

In the equatorial magnetosphere the amplitudes and the phases of the PPOs have been analyzed in detail by D. J. Andrews et al. (2019). The magnetic field fluctuations due to the PPOs have a northern and southern component rotating with independent and drifting phases, and the three magnetic field components can additionally be phase-shifted from each other. The amplitudes and the ratio of the northern and southern fluctuations change over Cassini's mission from a southern dominance before the vernal equinox in 2009 August to a northern dominance afterward. The PPO fluctuations also possess local time (LT)

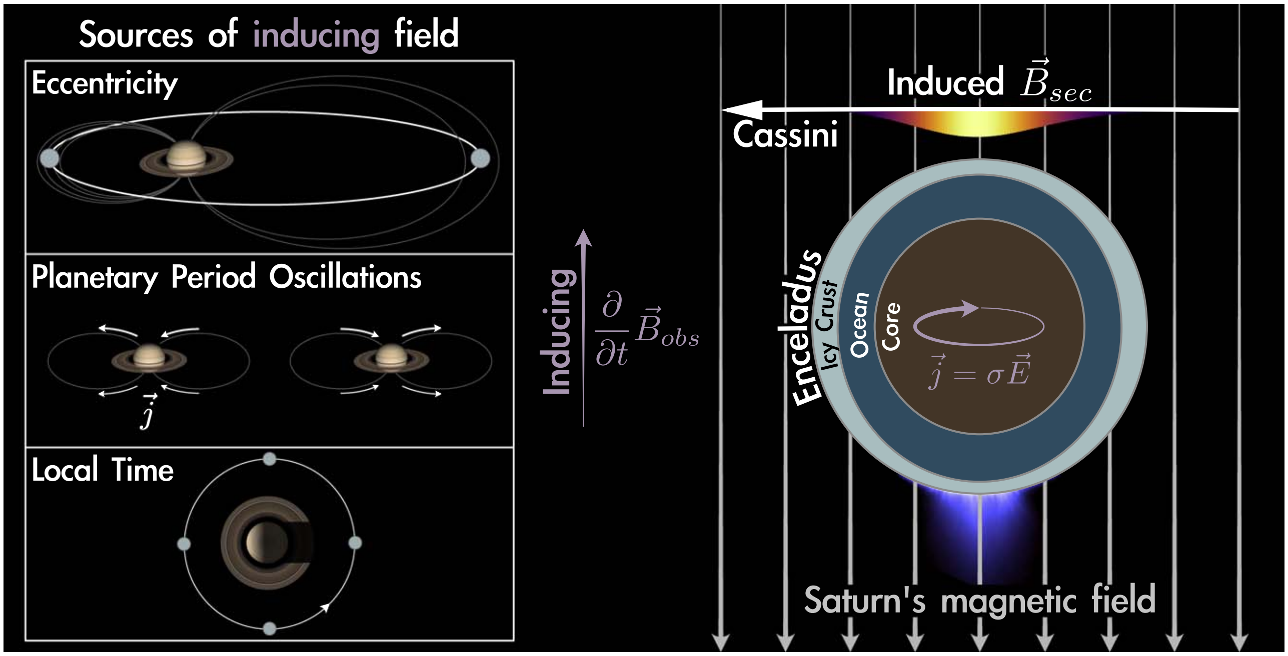

Figure 1. Sketch of induction processes at Enceladus. On the left-hand side, three mechanisms generating time-periodic, inducing fields around Enceladus are shown. On the right-hand side, Enceladus's three main layers, i.e., its icy crust, conducting ocean, and conducting core, are displayed. Within the conductive layers electric currents and induced magnetic fields are generated. The north polar region is particularly well suited for induction studies through spacecraft flybys (Plumes: Image Credit NASA).

and radial distance variability. Based on the analysis of G. Provan et al. (2009) and D. J. Andrews et al. (2019), we reconstruct the spatiotemporal structure of Saturn's PPO magnetic field over the Cassini mission time. The inducing fields due to PPO variability occur with the synodic rotation period of Saturn as observed in the rest frame of Enceladus. At the orbit of Enceladus they can have amplitudes of up to 1–2 nT. The associated period of the magnetic field variability is the synodic rotation period of Saturn as observed in the rest frame of Enceladus $T_{\mathrm{synodic}} = 0.6483$ days based on Saturn's nominal rotation period of $T_S = 0.4401$ days. The synodic rotation period associated with the PPO is not constant but changes for the northern and southern PPOs by a few percent over the Cassini mission, respectively. In the following, when we refer to the the PPO period variability, this effect is included in our calculations of the inducing phases.

2.3. Solar Wind, Saturn's Seasons, and Plasma Loading from Enceladus's Plumes

Saturn's magnetospheric field deviates from azimuthal symmetry largely because of the solar wind. The stresses of the solar wind on Saturn's magnetosphere generate Saturn LT and seasonal dependences. LT variability in Saturn's magnetosphere has been studied by A. M. Sorba et al. (2019), who find the lowest equatorial field strength for the night-into-the-dawn sector of the magnetosphere and the largest field strength on the noon-into-the-dusk sector. Saturn's large obliquity of 26.7° causes seasonal variability, which breaks the north–south symmetry in Saturn's magnetosphere described as a warped current sheet and magnetotail (C. S. Arridge et al. 2008). The magnetosphere also experiences compressed and expanded states (C. J. Martin & C. S. Arridge 2019). These results, however, have primarily been derived for the middle and outer magnetosphere. Saturn's magnetosphere is populated with mostly water group ions stemming from ionization of the neutral gas from Enceladus's plumes (M. K. Dougherty et al. 2006; R. L. Tokar et al. 2006; J. Saur et al. 2008; M. M. Hedman et al. 2013). The associated radial transport of plasma generates a plasma sheet with associated magnetic field perturbations referred to as a current sheet (e.g., C. S. Arridge et al. 2008; C. J. Martin & C. S. Arridge 2019; A. M. Sorba et al. 2019).

The LT variability causes time-periodic magnetic fields at the orbital period of Enceladus. The variability with the seasons is associated with Saturn's orbital period of 29 yr, which, however, does not generate measurable induction effects owing to its very slow variability. Compressional effects of Saturn's magnetosphere due to variable ram pressure of the solar wind have a stochastic component and a possible variability with the solar rotation period of 27 days (M. Seufert et al. 2011). Variability of the plumes might cause variability of the magnetic environment on the orbital period of Enceladus. In Section 3.2 and onward we therefore investigate which variability can be extracted from the Cassini data at radial distances of Enceladus's orbit stemming from magnetospheric effects. The associated period of the magnetic field variability is, in the case of the LT variability, not Enceladus's sidereal orbital period T_{orbit} but slightly modified owing to Saturn's orbit around the Sun with a period of about 29.5 yr. The resultant synodic orbital period is, however, only 0.01% larger. In the following, when we refer to the orbital period variability, this effect is included in our calculations of the inducing phases.

3. Analysis of Periodic Background Fields within Cassini Data

In this section we analyze magnetic field measurements taken near the orbit of Enceladus to characterize the time-

Table 1
Overview of Enceladus Flybys within $15R_E$

Name	Time C/A (UTC)	Dist C/A (R_E)	LT (hr)	$B_{z,ind}$ (nT)	Properties
En00	2005-02-17 03:30:28.5	5.90	22.6	2.1	distant
En01	2005-03-09 09:08:02.5	2.92	17.0	4.4	distant, south
En02	2005-07-14 19:55:20.5	1.63	17.1	2.3	medium distant, north–south
En03	2008-03-12 19:06:11.5	1.17	23.2	−0.2	close, north–south
En04	2008-08-11 21:06:18.5	1.17	22.7	3.8	close, north–south
En05	2008-10-09 19:06:39.5	1.08	22.5	4.7	close, north–south
En06	2008-10-31 17:14:51.5	1.64	22.5	4.8	medium distant, north–south
En07	2009-11-02 07:41:57.5	1.40	10.9	0.5	medium distant, south
En08	2009-11-21 02:09:56.5	7.20	3.7	4.3	distant, south, Alfvén wing
En09	2010-04-28 00:10:17.5	1.36	9.3	4.8	close, south polar
En10	2010-05-18 06:04:39.5	2.68	3.9	0.7	distant south
En11	2010-08-13 22:30:49.5	10.9	3.6	−1.9	distant south, Alfvén wing
En12	2010-11-30 11:53:59.5	1.15	8.8	1.75	close, north polar
En13	2010-12-21 01:08:26.5	1.16	8.8	1.2	close, north polar
En14	2011-10-01 13:52:25.5	1.35	23.8	2.3	close, south polar
En15	2011-10-19 09:22:11.5	5.78	23.8	2.7	distant, south polar
En16	2011-11-06 04:58:53.5	2.92	23.8	3.0	distant, south
En17	2012-03-27 18:30:08.5	1.25	0.6	4.5	close, south polar
En18	2012-04-14 14:01:37.5	1.25	0.6	4.6	close, south polar
En19	2012-05-02 09:31:28.5	1.25	0.5	4.7	close, south polar
En20	2015-10-14 10:41:28.5	8.15	12.2	−4.3	distant, north
En21	2015-10-28 15:22:41.5	1.16	20.8	1.2	close, south polar

Note. Rows provide flyby name, time of closest approach C/A, distance of Cassini to the center of Enceladus at C/A, Saturn LT, maximum induction response $B_{z,ind}$ due to eccentricity, and properties of the flyby.

variable, periodic magnetic fields Enceladus is exposed to. We consider all magnetic field measurements[3] during the Enceladus flybys and orbit crossings. The fields have been measured with the dual technique magnetometer system on board the Cassini orbiter with a sensitivity of 0.0488 nT (M. K. Dougherty et al. 2004). This set of magnetic field measurements can be subdivided into Enceladus flybys, defined by a closest approach within 15 Enceladus radii ($R_E = 256.6$ km), and crossing of Enceladus's orbit where the crossing occurred at distances larger than $15R_E$. The crossings are Cassini's passages of the radial distance of Enceladus's elliptical orbit within ±1.0° latitude. As the elliptical orbit is precessing with a period of 2.9 yr (R. Jacobson 2022), we need to determine the current orbit of Enceladus for each crossing individually. Using these categories, we find 21 Enceladus flybys and 50 orbit crossings. An overview with properties of the flybys is listed in Table 1. All geometrical/astrometric calculations are performed using SPICE.[4]

For the orbit crossings and the flybys we determined the background magnetic field at the crossing of Enceladus's slowly precessing elliptical orbit or at the time of the closest approach for the flybys. Close to Enceladus the magnetic field is perturbed owing to the plasma interaction with Enceladus's plumes. We therefore took the magnetic field measurements from regions upstream of the flyby not contaminated by the plasma interaction and extrapolated them to the position of closest approach. The downstream region often contains perturbations still stemming from plasma wake effects. This procedure leaves us with 71 values of magnetic field measurements to characterize the time-variable background field around Enceladus at different times throughout the Cassini mission from 2005 to 2015. In the following we perform this characterization in two steps. We first estimate the predictability of the time-varying background field due to the eccentric orbit and the PPOs, which are both known to be present with certainty based on previous studies and are thus zero-free-parameter predictions (Section 3.1). In the subsequent subsections we characterize the remaining residual time-variable components (Sections 3.2–3.5).

3.1. Basic Separation in Time-constant and Time-variable Fields

For the 71 background magnetic field values $\mathbf{B}_{obs}$, we subtracted Saturn's internal magnetic field $\boldsymbol{B}_{\rm Saturn}$ taken at Enceladus's average distance from Saturn $r_{av} = 238{,}040$ km (calculated with SPICE) and at colatitude $\Theta = 90°$. This field corresponds to the average background field at Enceladus stemming from Saturn's internal field. The internal field is calculated based on the Gauss coefficients from M. K. Dougherty et al. (2018). The resultant magnetic field perturbations of all flybys and orbit crossings

$$\Delta\boldsymbol{B}_{\rm obs} = \boldsymbol{B}_{\rm obs} - \boldsymbol{B}_{\rm Saturn}(r = r_{av}, \Theta = 90°) \qquad (1)$$

as a function of time are shown in Figure 2 as blue circles for the three magnetic field components in Saturn IAU planetocentric spherical coordinates, i.e., B_r, B_Θ, and B_ϕ. The magnetic field exhibits time variability between flybys and orbit crossings on the order of 10–20 nT. In order to understand

[3] A total of 294 PDS data products from the Cassini Orbiter MAG Calibrated Summary 1 Sec Averages V2.0 Data Set were used for the analysis performed within this work. The products contain magnetic field data in KRTP coordinates from 2004 May 14 to 2017 September 12 (M. K. Dougherty et al. 2019).

[4] This work used the following SPICE KERNELS: leaf seconds: naif0012.tls from https://naif.jpl.nasa.gov/pub/naif/generic_kernels/lsk/; ephemerides (spk kernels): reconstructed trajectories from 2020 January 28; filenames "200128RU_SCPSE_*.bsp" from https://naif.jpl.nasa.gov/pub/naif/CASSINI/kernels/spk/.

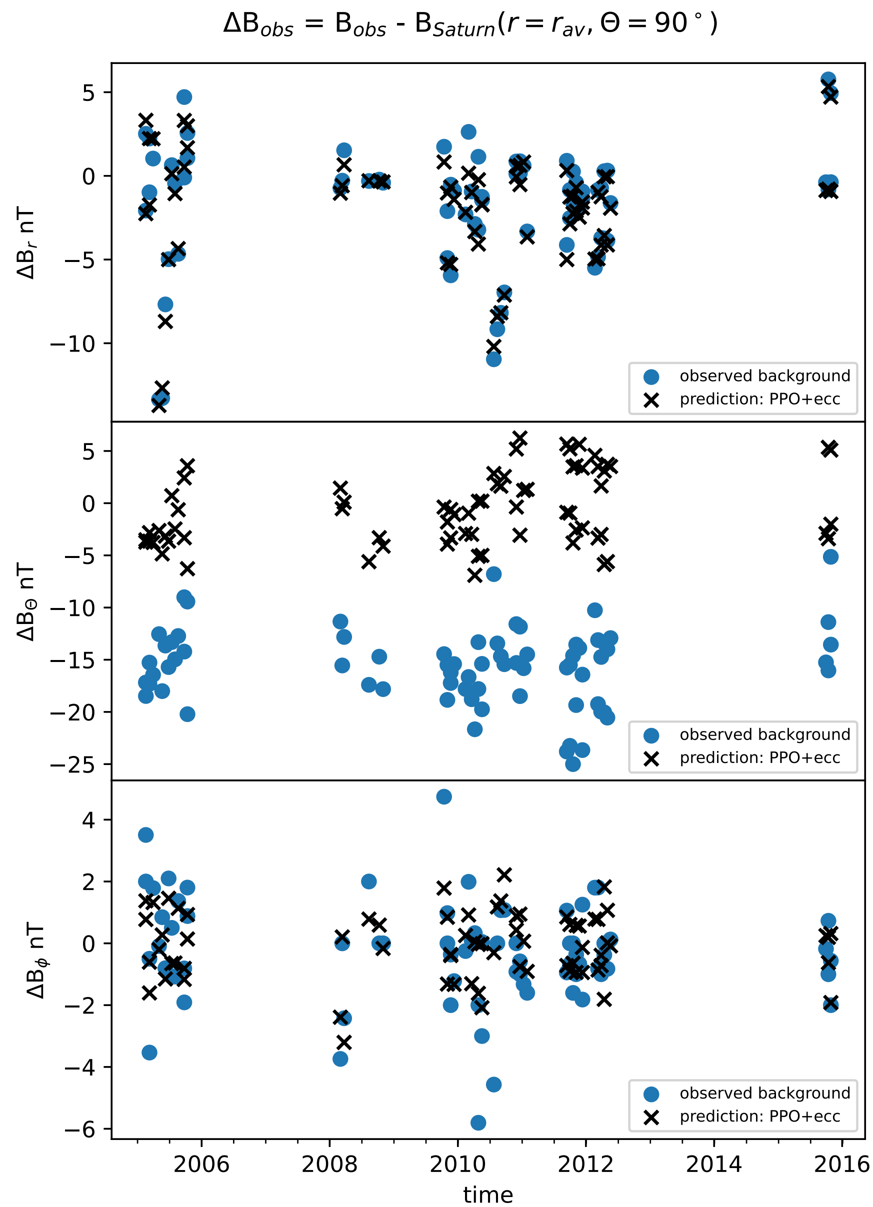

Figure 2. Predictability study of background magnetic field perturbations observed during Enceladus's flybys and crossings of Enceladus's orbit by the Cassini spacecraft. Circles show observed magnetic field perturbations calculated as deviation from Saturn's average time-constant internal field taken at Enceladus's average distance from Saturn $r = r_{\rm av}$ and colatitude $\Theta = 90°$. Crosses show predicted background field perturbations due to Enceladus's eccentric orbit at the location of the crossings/flybys and the PPO fields.

whether and how much of the variability is predictable, we calculate the expected magnetic field perturbations from the eccentric orbit within Saturn's internal magnetic field, which we refer to as $\boldsymbol{B}_{\rm ecc}(r, \Theta, \Phi)$, and those from the PPO, which we refer to as $\boldsymbol{B}_{\rm PPO}(r, \Theta, \Phi)$. The deviation between the predicted field perturbations and the observed field perturbations $\Delta\boldsymbol{B}_{\rm obs}$ from Equation (1) is defined as

$$\Delta\boldsymbol{B}_{\rm res} = \Delta\boldsymbol{B}_{\rm obs} - \boldsymbol{B}_{\rm ecc} - \boldsymbol{B}_{\rm PPO}. \tag{2}$$

We refer to them in the following as "residual magnetic fields," i.e., they describe observed variability in addition to our predicted PPO variability and the variability due to Enceladus's eccentric orbit within Saturn's magnetic field.

To characterize the quality of our predictions, we determine linear correlation coefficients c_r between predicted and observed background fields (listed in Table 2). We find that for B_r both the effects of Enceladus's eccentric orbit and the PPO signals predict the observed perturbations to a high degree with a correlation coefficient of 0.99. The strong predictability is visible in the top panel of Figure 2. For the radial component, the dominant part of the variability, however, does not describe the real magnetic variability Enceladus is exposed to. The dominant part of the measured variability in B_r stems from the sensitivity of the B_r component to the latitudinal position Θ where the measurements have been taken and not from the radial variability of Enceladus's orbit. This can be specifically seen with a dipole expansion shown in Equation (A7) in Appendix A, which has only a longitudinal dependence $\delta\Theta$ but no radial dependence δr. Cassini never crosses at Enceladus's orbital distance simultaneously Enceladus's exact orbital plane, i.e., at $\Theta = 90°$ with Enceladus's inclination being $i = 0.0°$. To

Table 2
Quality of Predicted versus Observed Background Magnetic Fields

Origin	Component	c_r	Mean of $\Delta\boldsymbol{B}_{\rm res}$ (nT)	c_σ of $\Delta\boldsymbol{B}_{\rm res}$ (nT)
PPO	B_r	0.079	−1.63	3.62
PPO	B_Θ	0.21	−15.87	3.59
PPO	B_ϕ	0.671	−0.19	1.25
ecc	B_r	0.975	0.22	0.81
ecc	B_Θ	0.631	−14.89	2.99
ecc	B_ϕ	0	−0.282	1.69
PPO, ecc	B_r	0.987	0.251	0.60
PPO, ecc	B_Θ	0.658	−15.1	2.96
PPO, ecc	B_ϕ	0.671	−0.190	1.25
PPO, ecc, season	B_r	0.987	0.0	0.58
PPO, ecc, season	B_Θ	0.767	0.0	2.42
PPO, ecc, season	B_ϕ	0.695	0.0	1.21
PPO, ecc, LT	B_r	0.987	0.0	0.59
PPO, ecc, LT	B_Θ	0.746	0.0	2.48
PPO, ecc, LT	B_ϕ	0.744	0.0	1.13
PPO, ecc, anomaly	B_r	0.987	0.0	0.59
PPO, ecc, anomaly	B_Θ	0.698	0.0	2.62
PPO, ecc, anomaly	B_ϕ	0.707	0.0	1.19
PPO, ecc, CS	B_r	0.987	0.0	0.59
PPO, ecc, CS	B_Θ	0.655	0.0	2.77
PPO, ecc, CS	B_ϕ	0.700	0.0	1.21
PPO, ecc, CS+LT	B_r	0.987	0.0	0.59
PPO, ecc, CS+LT	B_Θ	0.746	0.0	2.44
PPO, ecc, CS+LT	B_ϕ	0.748	0.0	1.12
PPO, ecc, CS+LT +season	B_r	0.99	0.0	0.54
PPO, ecc, CS+LT +season	B_Θ	0.86	0.0	1.88
PPO, ecc, CS+LT +season	B_ϕ	0.79	0.0	1.00

Note. First column: effects considered in predicting background field, where PPO stands for planetary period oscillation, ecc for eccentricity, season for the seasonal dependence, LT for Saturn local time, anomaly for phase with respect to the pericenter of the orbit, and CS for current sheet effects. The second column names magnetic field components. The third to fifth columns display correlation coefficient c_r between predicted and observed fields, average difference, and centralized rms difference, i.e., the standard deviation c_σ between predicted and observed magnetic field (for further details, see the main text).

illustrate the point further, let us choose exemplary values at colatitudes $90^\circ \pm 0.1^\circ$ or $90^\circ \pm 1.0^\circ$, which correspond to heights of $\pm 1.6 R_{\rm E}$ and $\pm 16 R_{\rm E}$ above or below Enceladus's spin plane, respectively. Measurements at such height differences would cause a variability of ± 1.2 and ± 11.7 nT between the opposite colatitudinal sides at $90^\circ \pm 0.1^\circ$ and $90^\circ \pm 1.0^\circ$, respectively. This effect is the primary cause of the variability in the top panel of Figure 2. Only a small contribution of real variability in B_r at the location of Enceladus stems from the eccentricity (it is included in the higher-degree terms in Equation (A4), see also Section 2.1). Another small contribution stems from the PPO fields, which can assume values on the order of 1–2 nT. For the B_Θ component the effect of Cassini crossing the elliptical orbit of Enceladus somewhat above or below the moon´s real path has, however, only very small effects. This can be seen specifically from a dipole expansion shown in Equation (A8) in Appendix A, which has only a radial dependence δr but no longitudinal dependence $\delta\Theta$. A flyby at $90^\circ \pm 0.1^\circ$ or $90^\circ \pm 1.0^\circ$ corresponds to only ± 0.04 nT or ± 0.4 nT variability, respectively, using the full expansion of Equation (A5). Due to the azimuthal symmetry, the internal component of Saturn's field $B_\phi = 0$ is not affected by radial or latitudinal effects (see Equation (A6)). Thus, B_Θ and B_ϕ are only minimally or not at all affected by the "height of the crossing effect."

The residual difference between observed and predicted fields shown in Figure 2 can be quantified by the mean values of the residual fields $\langle \Delta\boldsymbol{B}_{\rm res}\rangle$, with angle brackets denoting the mean over all 71 data points. The standard deviation from the mean residual fields is $c_\sigma = \langle (\Delta\boldsymbol{B}_{\rm res} - \langle \Delta\boldsymbol{B}_{\rm res}\rangle)^2\rangle^{0.5}$. The latter two quantities are given in the two last columns of Table 2. For B_Θ we find a correlation coefficient of 0.66 between observed field perturbations and the predicted fields owing to eccentricity and PPO effects. For the Θ component the PPO plays a larger role compared to the radial component (Table 2). Most notably, there is a large average offset of about −15 nT of predicted and observed field (Figure 2 and Table 2). An average offset of about −15 nT is due to current sheet effects in Saturn's magnetosphere as reported previously (e.g., M. K. Dougherty et al. 2018). This average offset and its standard deviation of about 3 nT will be analyzed and separated into time-constant and time-variable contributions in the following subsections. For the B_ϕ component the correlation coefficient c_r is 0.67. It only has contributions from PPO and none from the eccentric orbit since Saturn's internal magnetic field has no known deviation from azimuthal symmetry and no B_ϕ component (M. K. Dougherty et al. 2018).

In summary, we see that Enceladus's orbital eccentricity and the PPO fields are predictable periodic magnetic field perturbations, which capture observed magnetic field perturbations Enceladus is exposed to with correlation coefficients of 0.66 and larger within a zero-free-parameter fit. In the next subsections we investigate the possible causes behind the residual magnetic field perturbations. Therefore, we first investigate additional effects such as LT, seasonal, and other effects independently, and afterward we study their cross-correlations.

Our analysis and characterization of the time-periodic fields near Enceladus is a two-step procedure. The first step was executed in this subsection, i.e., we investigated the difference of the observed background fields compared to those from the eccentricity and the PPO fields called residuals. In the next subsections we will perform the second step and investigate additional time-periodic variability remaining in the residuals. This two-step procedure has the advantage that it filters in the first step the "height of the crossing" effects. While we can investigate the magnetic field at the exact radial position of Enceladus's eccentric orbit during each crossing, Cassini crosses always a bit above or below Enceladus's true orbit (see discussion two paragraphs further up). Working with the residuals filters this effect because we subtract Saturn's internal field at the corresponding height above the equator through $\boldsymbol{B}_{\rm ecc}$ in Equation (2) and thus minimize/suppress the "height of the crossing" effect. Now we continue with the second step and analyze the residuals.

3.2. Seasonal Dependence of Residual Perturbations

In Figure 3, we display the seasonal dependence of the residual fields from all flybys and orbit crossings as blue circles. Saturn's vernal equinox was on 2009 August 11. We refer to the time dependence over the Cassini mission also as seasonal dependence. Saturn's spin axis is tilted by 26.7°. This

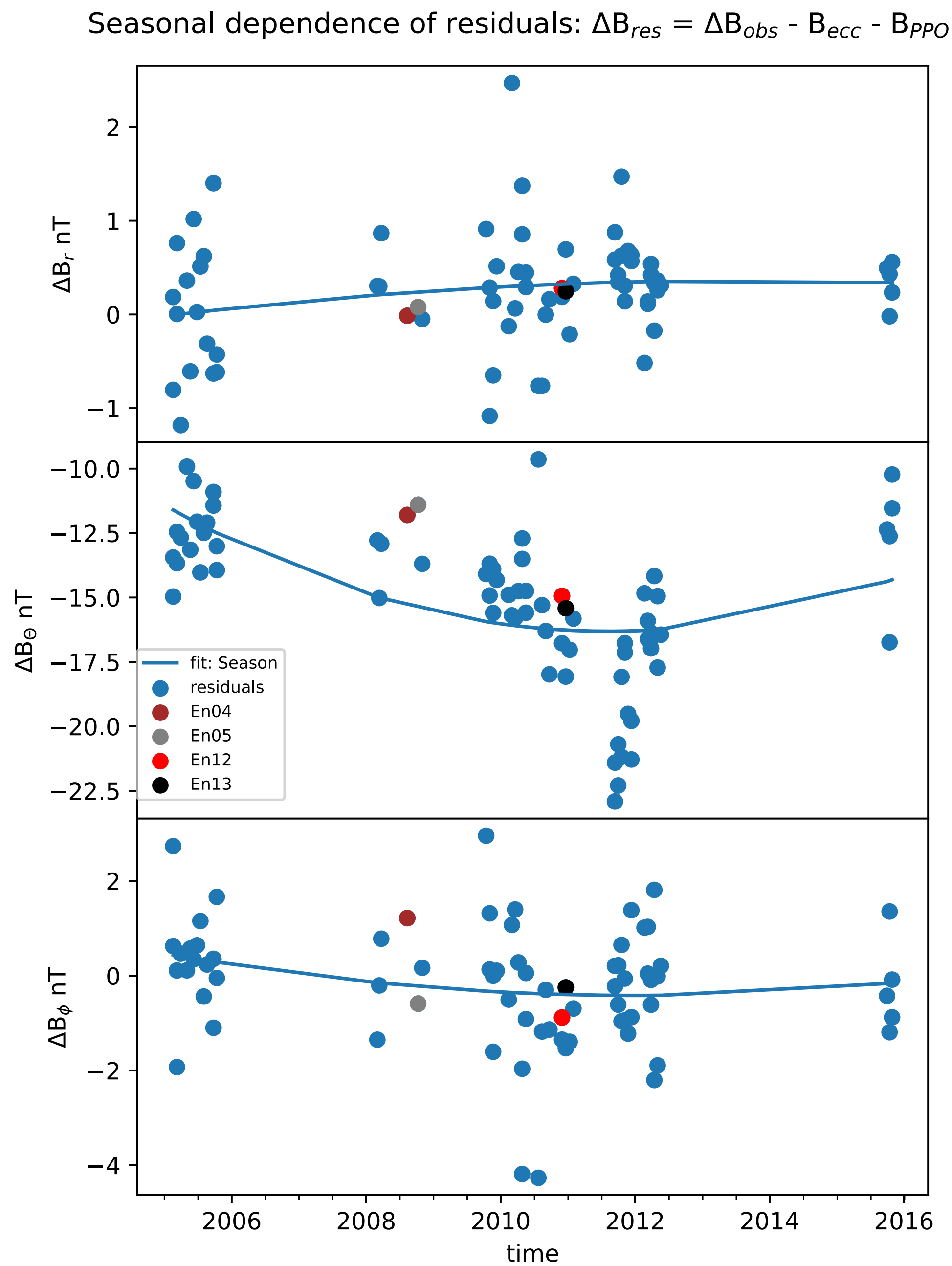

Figure 3. Time dependence of residual magnetic fields stemming from the difference of the observed background magnetic fields and those of Saturn's internal field and the PPOs (see Equation (2)). The observed magnetospheric background fields are for either flybys or crossings of Enceladus's eccentric orbit shown as blue circles. Values for selected flybys are shown in differently colored circles. The blue curves display polynomial fits to the data.

tilt changes Saturn's magnetospheric field over a Saturn year with a period of 29.5 yr. The colored circles highlight particular flybys to be discussed in the following sections. The solid blue lines represent quadratic polynomial fits to the data points. While there seems to be only a very mild seasonal dependence for the radial component and the azimuthal component, a stronger variability appears to exist for the latitudinal component. The fit captures first of all the offset of about -15 nT discussed in the previous subsection due to plasma and current sheet effects. The variability over the seasons shows that the largest decrease of the latitudinal component and thus also approximately of the magnetic field magnitude occurs around equinox. Considering the seasonal dependence increases the correlation coefficient c_r to 0.76 for the latitudinal component and to 0.7 for the azimuthal component, while the radial component with $c_r = 0.99$ is nearly perfectly correlated (see Table 2). The standard deviations c_σ of the residuals from the seasonal fit are 2.42 nT for the latitudinal component and 0.58 and 1.21 nT for the radial and azimuthal components, respectively (Table 2).

3.3. Local Time Dependence of Residual Perturbations

In Figure 4, we display the residual magnetic fields, defined by Equation (2), as a function of Saturn's LT. The data show systematic trends overlaid by variability not controlled by LT. To extract the systematic variability, we fit the data to a three-

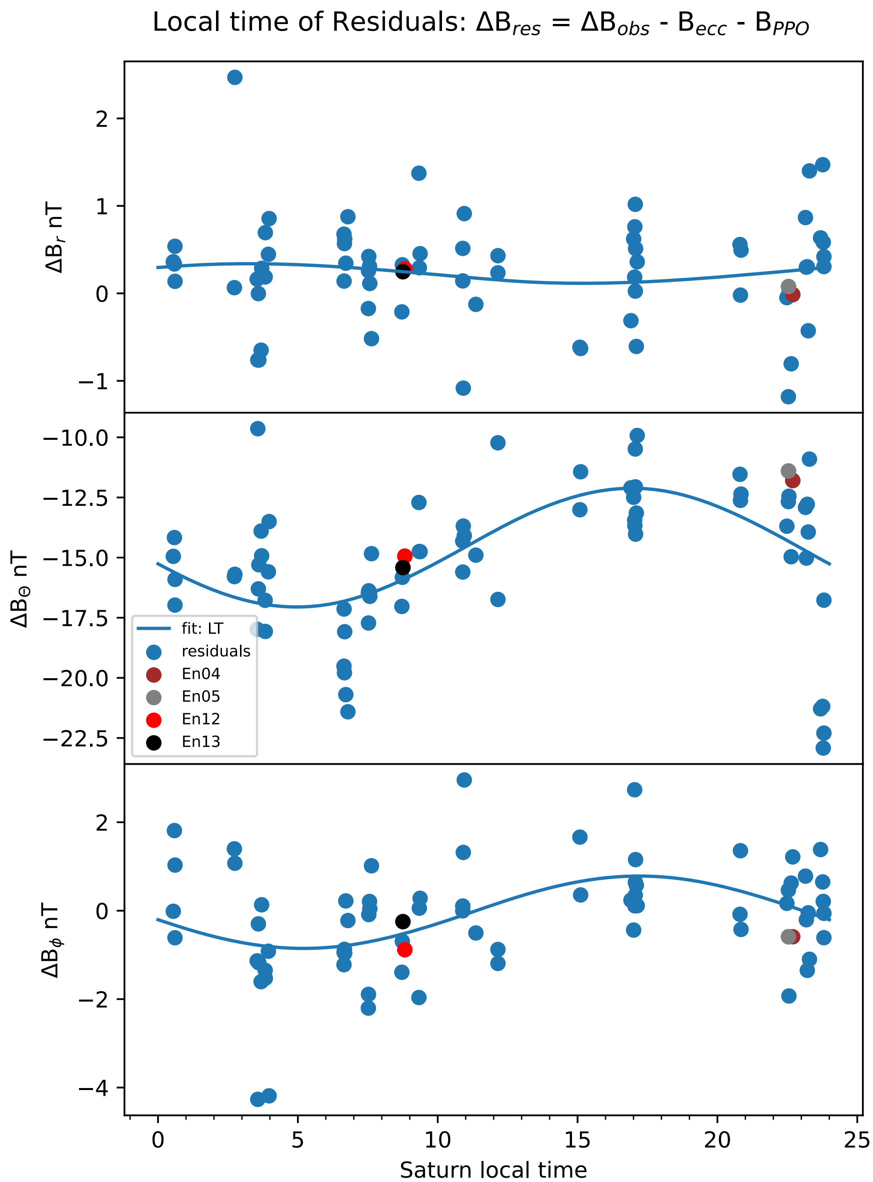

Figure 4. Saturn LT dependence of residual magnetic fields stemming from the difference of the observed background magnetic fields and those of Saturn's internal field and the PPOs (see Equation (2)). The observed magnetospheric background fields are for either flybys or crossings of Enceladus's eccentric orbit. Values for selected flybys are shown as differently colored circles. The blue curves display sinusoidal fits to the data.

parameter function of the form $B_0 + B_{\cos}\cos(2\pi/24(\varphi_0 + \varphi))$ for each of the three magnetic components with the fit parameters B_0, $B_{\cos}$, and φ_0. The angle φ represents either LT or anomaly depending on the fit. The results of the fits are shown as blues lines in Figure 4, and the values for the parameters of the dominating inducing component, i.e., B_Θ, of the residual fields are listed in Table 3. The largest variability with also the largest systematic offset of about −15 nT occurs in the latitudinal component as captured by the fit (see Table 3). The offset is ascribed to Saturn's magnetodisk (C. S. Arridge et al. 2008; A. M. Sorba et al. 2019). The latitudinal component appears to show a periodic variability with amplitude of 2.5 nT as a function of LT where the minimum occurs at 4.9 hr, i.e., between the night and the dawn side of the magnetosphere. This roughly fits with the trend of the LT variability seen at larger radial distances by A. M. Sorba et al. (2019), where the weakest magnetic field is reported after midnight toward the morning sector of Saturn's magnetosphere.

Table 3
Fit Coefficients to Residual Fields of the Form $B_0 + B_{\cos}\cos(\varphi_0 + \varphi)$

Dependence	Component	B_0 (nT)	$B_{\cos}$ (nT)	φ_0	B_{CS} (nT)
LT	B_Θ	−14.6	−2.5	−4.9 hr	
Anomaly	B_Θ	−15.3	−1.9	0.1 phase	
CS, LT	B_Θ	−14.7	−2.2	−4.4 hr	−0.9
CS, LT, equinox	B_Θ	−14.1	−1.5	−6.7 hr	−1.2

Note. The values B_0, $B_{\cos}$, and φ_0 are fit parameters. The angle φ represents either LT or anomaly depending on the fit. LT is measured in hr. The mean anomaly we measure in phase, i.e., 0−1, corresponds to 0°−360°. The fit parameter B_{CS} describes the value with which the magnetic field strength due to eccentricity is reduced by the current sheet.

Including possible LT variability as a predictive component increases the correlation coefficient to 0.75 for the latitudinal component and to 0.75 for the azimuthal component (Table 2).

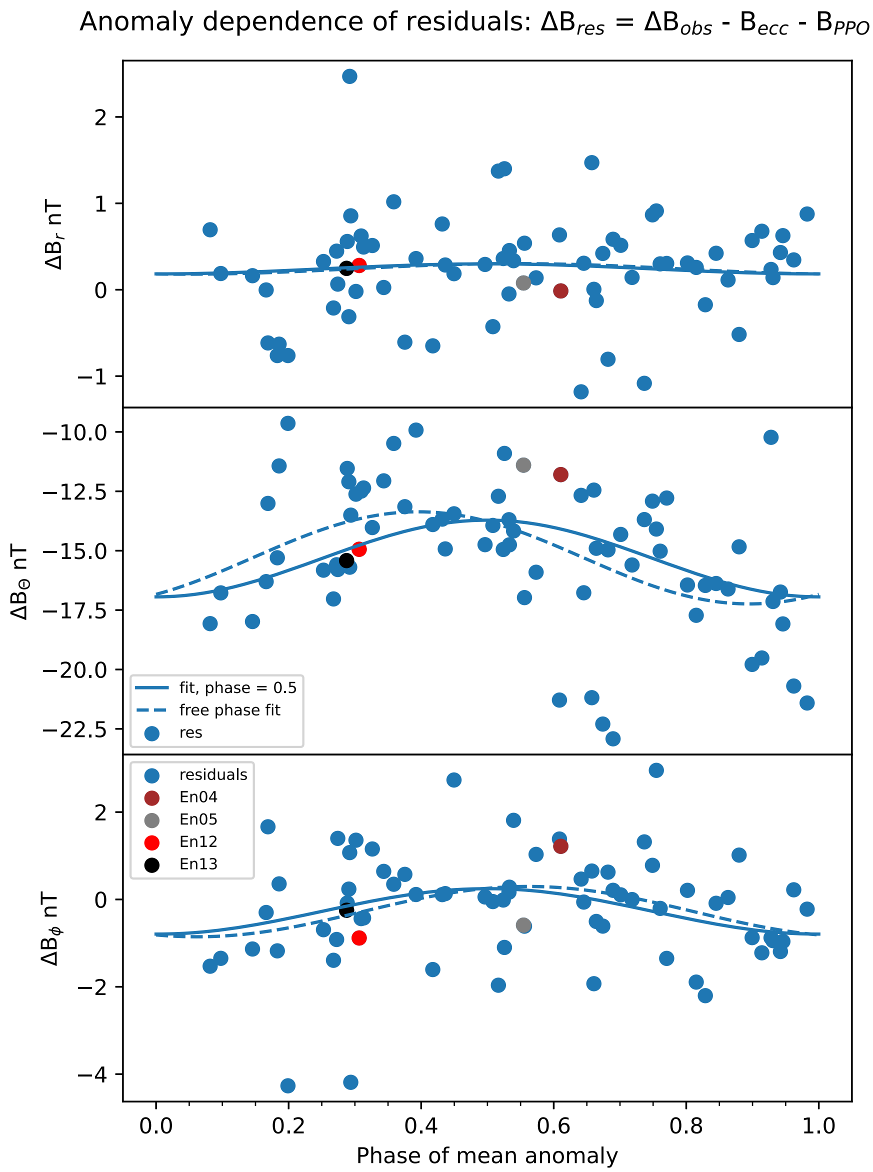

Figure 5. Anomaly dependence of residual magnetic fields stemming from the difference of the observed background magnetic fields and those of Saturn's internal field and the PPOs (see Equation (2)). The residual fields are displayed as a function of the mean anomaly of the position of measurements by the Cassini spacecraft for either flybys or crossings of Enceladus's eccentric orbit. The mean anomaly is the angle with respect to the location of Enceladus's pericenter before the flyby. The mean anomaly is expressed as phase with values lying between 0 and 1, which corresponds to 0° and 360°. Values for selected flybys are shown as differently colored circles. The blue dashed curves display sinusoidal fits to the data with a free phase shift. The blue solid curves represent a sinusoidal fit with the phase shift fixed such that the maxima are at phase = 0.5.

The remaining standard deviations for the radial, latitudinal, and azimuthal components are 0.59, 2.48, and 1.13 nT, respectively (Table 2).

3.4. Mean Anomaly Dependence of Residual Perturbations

In Figure 5 we display the dependence of the residual magnetic field perturbations as a function of the mean anomaly. The mean anomaly is the angle from the pericenter, here expressed as phase, i.e., phase = 0 corresponds to 0° and phase = 1 to 360°. The position of Enceladus's eccentric orbit is controlled by the mean anomaly, and the effect of the eccentricity within Saturn's internal field has already been included in the the analysis of the previous section. Here we search for additional variability controlled by the mean anomaly. We fit the data to a sinusoidal dependence with free phase shift shown as blue dashed lines. The fit parameters of the dominating inducing B_Θ component of the residual fields are listed in Table 3. The latitudinal and azimuthal components in Figure 5 show the largest values of the residual fields around phase = 0.4–0.5.

We now discuss the dependence on the mean anomaly, with two physical reasons in mind. First, the plume activity of Enceladus is maximum when Enceladus is near its apocenter (M. M. Hedman et al. 2013). The gas of the plumes will be ionized and larger plasma production rates emerge when the neutral density is larger. Larger production will cause a denser plasma sheet, which might have implications for the current sheet and its associated magnetic fields. However, the neutrals of the plume approximately revolve with Enceladus, but once ionized, the plasma is frozen into Saturn's magnetospheric field and will approximately rotate with Saturn. We thus calculate

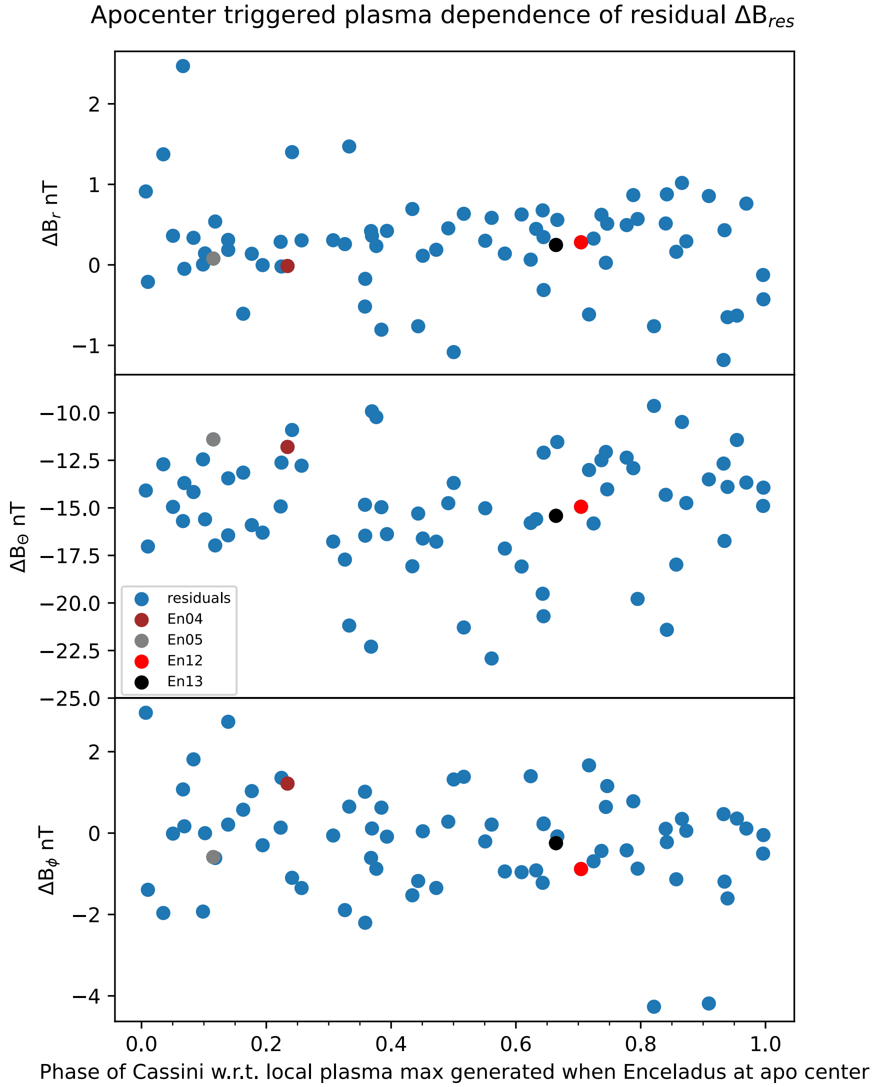

Figure 6. Search for dependence of residual magnetic fields on variability of Enceladus's plume activity. Residual magnetic fields are the difference between the observed background magnetic fields and those of Saturn's internal field and the PPOs. Enceladus plume activity is maximum when Enceladus is close to its apocenter (M. M. Hedman et al. 2013). The enhanced plasma production possibly generates a plasma pulse/max that will rotate with Saturn's magnetic field. The residual fields are displayed as a function of distance with respect to the location of a possible plasma pulse/max.

the phase of the magnetic field measurement with respect to a plasma pulse produced near the apocenter and then rotating with Saturn. The results are shown in Figure 6. Here we see no correlation, which does not appear to be very surprising since the plasma transport is expected to smear out densities along the Enceladus torus over torus timescales expected to be on the order of months.

The second reason for a correlation of the residual fields with the mean anomaly is that the mean anomaly is a measure for the orbital distance of Enceladus from Saturn, i.e., the mean anomaly is fully correlated with the eccentricity effects introduced earlier. The residuals in Figure 5 are smallest when Enceladus is approximately at its apocenter, i.e., farthest away from Saturn. We ascribe the residual dependence to current sheet effects. The residuals in our analysis are calculated with Saturn's internal field because it is a well-quantified field (M. K. Dougherty et al. 2018). The current sheet effects lead to magnetic fields that generally fall off less rapidly than those from the internal field only (except for the noon region of the magnetosphere). This effect is described by A. M. Sorba et al. (2019) in the middle and outer magnetosphere of Saturn and is also known from Jupiter's magnetosphere. Thus, the increase in ΔB_Θ around phase 0.5 most likely comes from a magnetospheric field that is larger at apocenter than predicted by the internal fields only. Thus, in turn, we can use the anomaly dependence measured here to constrain the current sheet magnetic field variability as a function of radial distance along the orbit of Enceladus. The current sheet dependence deep inside Saturn's magnetosphere has not been the focus of previous studies and cannot be quantitatively extracted from such studies (e.g., A. M. Sorba et al. 2019).

Now we first assume that the current sheet has no LT effects, or the LT effects are smeared out over the Cassini mission when measured as a function of the mean anomaly. We thus fix the phase to the apocenter of Enceladus's orbit such that the residual magnetic field components shown as dashed lines in Figure 5 have the largest values at phase = 0.5. The quality of the fit is similar to the fit without fixed phase (i.e., the solid line). The difference between both fits is given quantitatively in Table 2. For the B_Θ component, where the amplitudes are the largest, the correlation is 0.69 in the free-phase fit (labeled "PPO, ecc, anomaly" in Table 2) and 0.66 in the fixed-phase case (labeled "PPO, ecc, CS" in Table 2). The amplitudes of the sinusoidal fit in both cases are 1.9 and 1.6 nT, respectively. Using a Taylor expansion of the local magnetospheric field, this translates into a net average radial dependence near Enceladus's orbit due to current sheet effects of $B(r) \sim r^{-1.9}$ and $B(r) \sim r^{-2.0}$, respectively, and is expected to be associated with an Enceladus plasma torus (for further details on the Taylor expansion, see Appendix A).

3.5. Discussion and Conclusions from Analysis of Time-variable Fields

In Section 3.1 we first demonstrated that Enceladus's eccentric orbit and Saturn's PPO variations generate periodic variability, which we identified in Cassini measurements. In the three subsequent Sections 3.2, 3.3, and 3.4 we investigated residual dependencies of Enceladus's time-variable magnetic field environment as a function of seasons, LT, and mean anomaly. Unfortunately, these three quantities are not independent. The data coverage with respect to these three quantities is shown in Figure 7. The top panel of this figure shows that during certain parts of the Cassini mission Enceladus was observed at similar LTs. For example, before equinox in 2009 August only LTs from late afternoon to midnight are covered. In the middle panel of Figure 7, not all mean anomalies are covered equally well, e.g., during pre-equinox only anomaly phases from about 0.2 to 0.7 are covered. A number of clustered neighboring data points in this panel show a decrease with a nearly constant slope, which is due to the precession drift of Enceladus with a period of 2.9 yr (R. Jacobson 2022). The bottom panel of Figure 7 shows which regions in the LT–mean anomaly parameter space are observed by the Cassini measurements. Not all regions are covered evenly here as well.

Due to the uneven coverage of season, LT, and mean anomaly, the dependence of the residual time-variable magnetospheric field on these variables cannot be resolved uniquely with statistical methods. In the remainder of this work, we will use the statistically resolvable dependences and augment them with physically plausible scenarios. Therefore, we will apply the following strategy: (i) We assume that the anomaly dependence shown in Section 3.4 is primarily a current sheet effect, and we additionally allow for LT variabilities within each of the three time spans. (ii) We divide the Cassini mission into three seasonal data blocks with a "pre-equinox" phase, i.e., Cassini measurements before 2008, an "equinox" phase around equinox 2008–2010, and a "past equinox" phase for times later in the mission. We apply these assumptions subsequently and show the results and their differences for the primary inducting fields in Figure 8 and Table 2, as well as when we calculate induced fields in Section 5.

Considering point (i), we display in Figure 8 the results of our predicted background fields when we take into account PPO, the eccentricity, and LT effects within a current sheet as black crosses. The correlation coefficients c_r for the B_r, B_Θ, and B_ϕ components are 0.99, 0.75, and 0.75, respectively (see Table 2 and fit parameters of the dominating inducing B_Θ component of the residual fields in Table 3). Considering additionally the seasonal variability of point (ii), we show the predicted fields as pink, orange, and dark-red three-point stars for the phases labeled winter, equinox, and summer, respectively. The correlation coefficients c_r for B_r, B_Θ, and B_ϕ components are 0.99, 0.86, and 0.79 for the "equinox phase," respectively (see Table 2; for fit parameters of the dominating inducing component of the residual fields, see Table 3). The correlation coefficients are similar for the other seasonal phases. We highlight here the "equinox phase," as we will see in Section 5.2 that the most relevant flybys for our induction study occurred during this phase. These correlation coefficients and Figure 8 demonstrate that the background magnetic field Enceladus is exposed to can be predicted with relatively high accuracy when PPOs, the eccentricity, and magnetospheric effects are considered.

4. Induction Amplitudes and Phases

In order to calculate the possible electromagnetic induction response within Enceladus, we construct a simple spherically symmetric three-layer model of Enceladus's interior (see Figure 1). For each layer we assume a constant electrical conductivity even though Enceladus's ocean and core might exhibit salinity gradients (e.g., W. Kang et al. 2022a, 2022b; Y. Zeng & M. F. Jansen 2024) and deviations from radial symmetry. We neglect this complexity for the following reasons: (i) The actual stratification of the ocean, and hence the radial conductivity, depends strongly on its thermochemical state and fluid dynamics regime, both of which are currently unknown for icy moons, including Enceladus. (ii) Even if it is stratified, induction sounding with only one or a few long periods (i.e., when the diffusion length is larger than the thickness of the ocean) cannot resolve a radial conductivity dependency; thus, only the bulk averages are sounded at best. Hence, the usage of a constant conductivity (e.g., some effective-depth-averaged value) for the whole water column is currently the most justified approach. This reconciles with the conclusion found in previous work (e.g., S. D. Vance et al. 2021). (iii) Deviations from radial symmetry can be in principle included in modeling the induction response (e.g., E. B. Fainberg et al. 1990; A. V. Grayver et al. 2019; M. J. Styczinski et al. 2022), but for simplicity we work with a radially symmetric layering and discuss the effects of the uncertainty of the thickness of the crust for the north polar flybys.

Our model assumes a porous core permeated by water with a radius of 194 km (P. C. Thomas et al. 2016), an outer ice crust of only 5 km, and a resulting ocean thickness of 57 km. The average thickness of the crust after P. C. Thomas et al. (2016) is 21 km. However, near the south pole the thickness is expected to be only about 5 km or less. In the north polar region the thickness of the ice crust is estimated to be around 10 km, but can at some locations assume smaller values (e.g., G. Choblet et al. 2017; M. L. Cable et al. 2021). We apply our finite conductivity model for the north polar flybys only because these flybys are the most sensitive for the induction studies. We choose 5 km ice thickness as a lower limit case for the thickness of the crust in our radially symmetric model. The ice shell thickness at the north pole is not known with certainty, but a difference between 5 and 10 km leads to a difference in induction amplitude at the surface of Enceladus of $A = 0.94$ and 0.89, respectively, assuming that the induction amplitude at the top of the ocean is $A = 1$. On the other hand, we assume a core radius of 194 km based on P. C. Thomas et al. (2016), while R. S. Park et al. (2024) derive a larger core radius in the

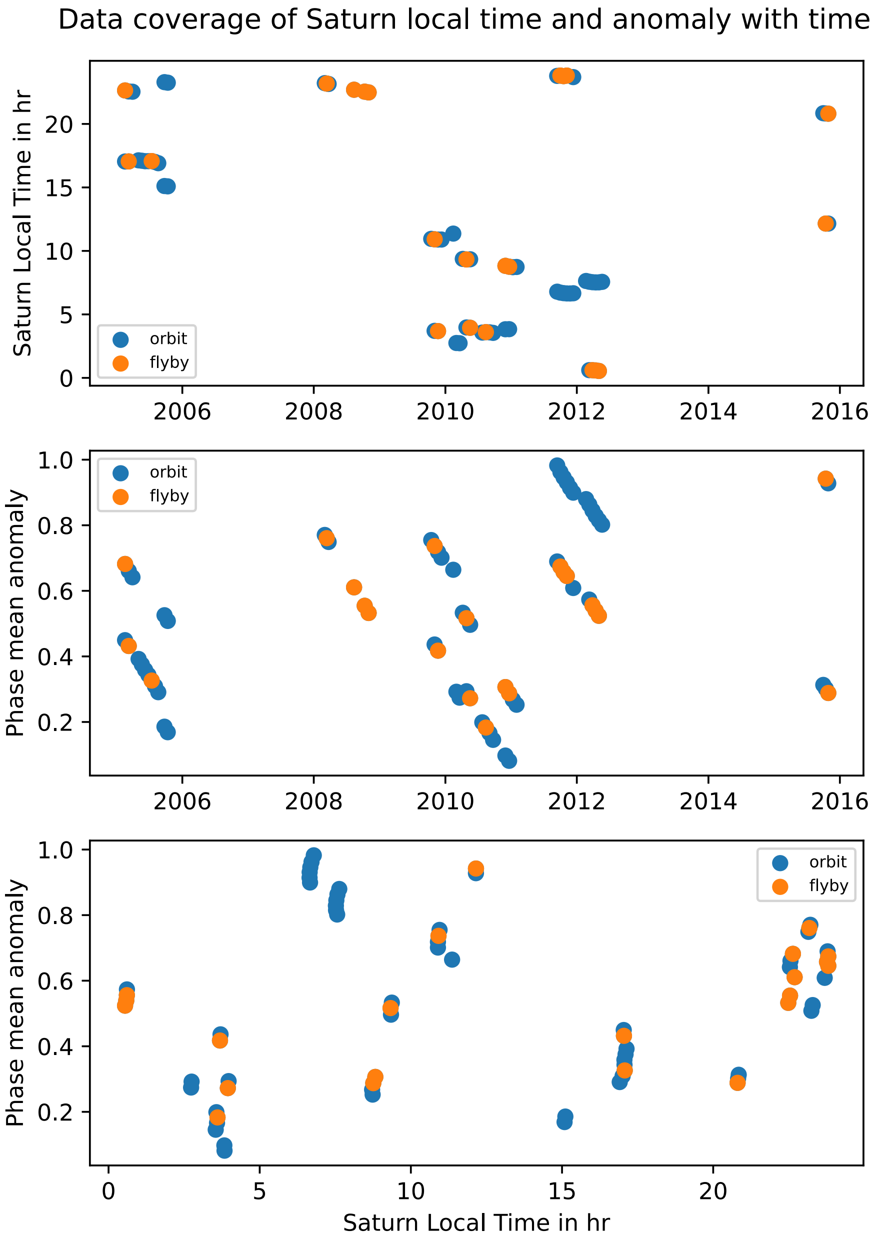

Figure 7. Data coverage with respect to time, Saturn LT, and mean anomaly during which Cassini measurements at Enceladus's flybys or at orbit crossings were obtained. Orange circles represent flybys, and blue circles represent orbit crossings.

range of 197–200 km. If we were to use a core radius of 200 km, this would lead to an enhanced induction response of $A = 0.47$ compared to 0.43 for a radius of 194 km, assuming $A = 1$ at the top of the core. These differences are small, and the uncertainties of the thickness of the ice shell or of the core are ultimately free parameters, but they translate effectively into an uncertainty of determining the induction amplitude A. As the conductivities of the ocean and the core are the much less constrained values compared to their extent, we treat the conductivities as free parameters and keep in mind that they include the uncertainty about their extent. In the following section we thus constrain the conductivities with the Cassini measurements and geophysical arguments.

The induction amplitude $A_n(\omega)$ and phase lag $\Phi_n(\omega)$ characterize the induced field from a finitely conducting body with respect to the induced field $\boldsymbol{B}_{\mathrm{sec},\infty}$ from a perfectly conducting body through

$$\boldsymbol{B}_{\mathrm{sec}}(t, \boldsymbol{x}) = A_n(\omega)\boldsymbol{B}_{\mathrm{sec},\infty}(t - \Phi_n(\omega)/\omega, \boldsymbol{x}) \tag{3}$$

with the inducing frequency ω (e.g., C. Zimmer et al. 2000; M. Seufert et al. 2011; M. J. Styczinski et al. 2022). They are related to the Q-response typically used in the geophysics community through the complex-valued relation $Q_n(\omega) = n/(n+1)A_n \exp(i\Phi_n)$, with n the degree of the spherical harmonics used to characterize the primary field (e.g., J. Saur et al. 2010; M. J. Styczinski et al. 2022; A. Grayver 2024). Since we assume that the time-variable field is spatially constant over the size of Enceladus, n is equal to 1 and $\boldsymbol{B}_{\mathrm{sec},\infty}(t, \boldsymbol{x})$ is a dipole field (e.g., C. Zimmer et al. 2000; J. Saur et al. 2010). In the following we study only degree $n = 1$ fields and thus omit the related subscript in the induction

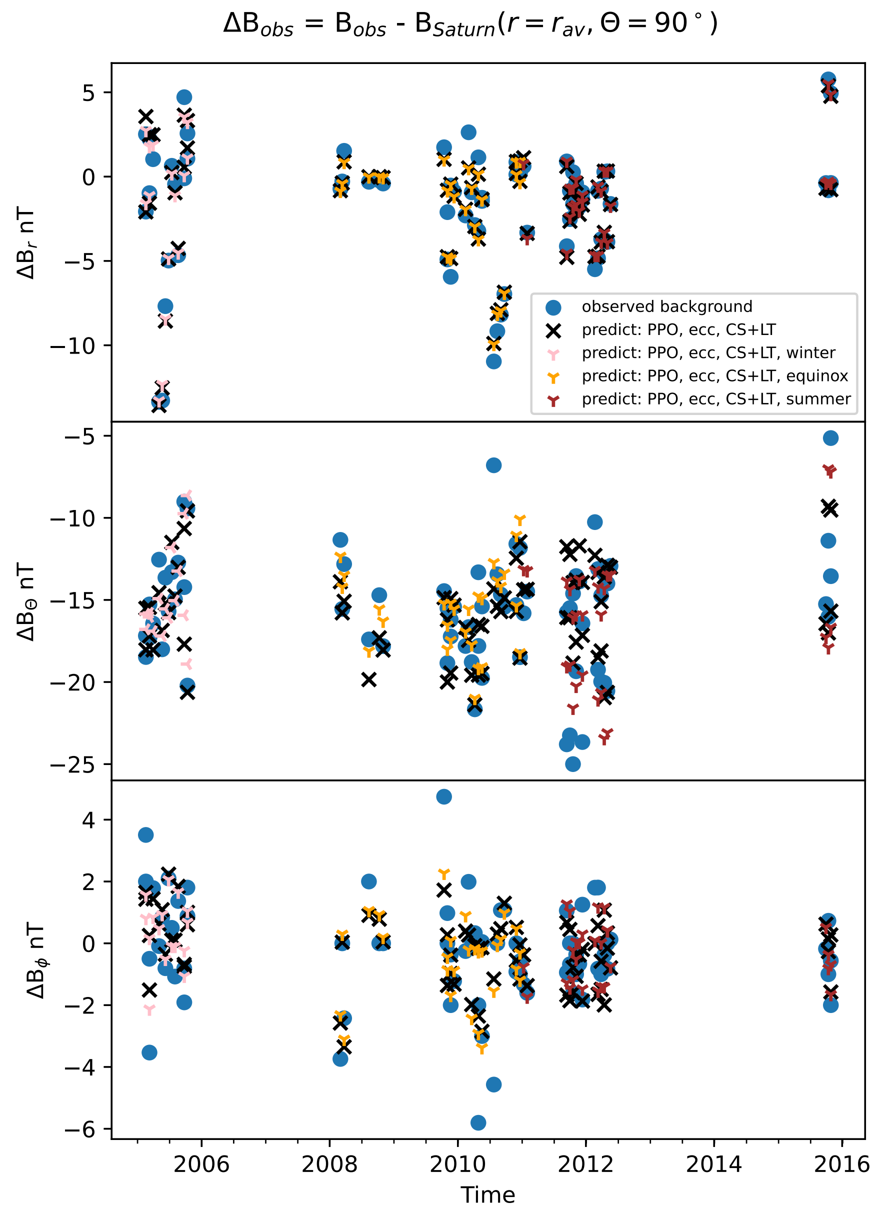

Figure 8. Improved predictability of primary inducing fields when PPO, eccentricity, and magnetospheric effects are included. Blue circles show residual background fields. Black crosses show fits including eccentricity, PPO, and LT effects within a current sheet. Orange, pink, and red symbols represent improved fits when the data are additionally binned into three seasonal blocks during the fitting process.

amplitude A_n and the phase Φ_n. The amplitudes A and the phases Φ can be calculated with standard methods for multilayered cases from the literature. These approaches vary in their mathematical details. Some solutions are based on modified spherical Bessel functions (N. Schilling et al. 2007; M. Seufert et al. 2011) or standard spherical Bessel functions (M. J. Styczinski & E. M. Harnett 2021; see also W. Parkinson 1983). We used an approach from E. B. Fainberg et al. (1990) and A. V. Kuvshinov (2008), which was implemented and tested in A. V. Grayver et al. (2019) as a part of a general 3D EM induction solver. Other implementations in K. Pěč et al. (1985) and S. P. Srivastava (1966) provide results in agreement with our solutions.

The results for the amplitudes A and phases Φ for our conductivity model are displayed in Figure 9. The two inducing frequencies/periods associated with the variability discussed in Sections 2 and 3 are the orbital period of Enceladus with $T_{\rm orbit} = 1.3702$ days (R. Jacobson 2022) and the synodic rotation period of Saturn as observed in the rest frame of Enceladus $T_{\rm synodic} = 0.6483$ days based on Saturn's nominal rotation period of $T_S = 0.4401$ days. The variability of the inducing magnetic field occurs at periods slightly shifted compared to $T_{\rm orbit}$ and $T_{\rm synodic}$ as discussed in Sections 2.1–2.3. These shifts have negligible effect on the precision of the induction amplitudes and phases required for this analysis. The amplitudes are calculated for the surface of Enceladus and thus include the falloff by r^{-3} within the crust.

We find that induction amplitudes are generally larger and the phase smaller for the shorter synodic period compared to the orbital period. For negligible ocean conductivity, a highly conductive core would lead to an induction response of about 0.5 with a small phase. The decrease to 0.5 mostly stems from the distance of the top of the core to the surface of Enceladus.

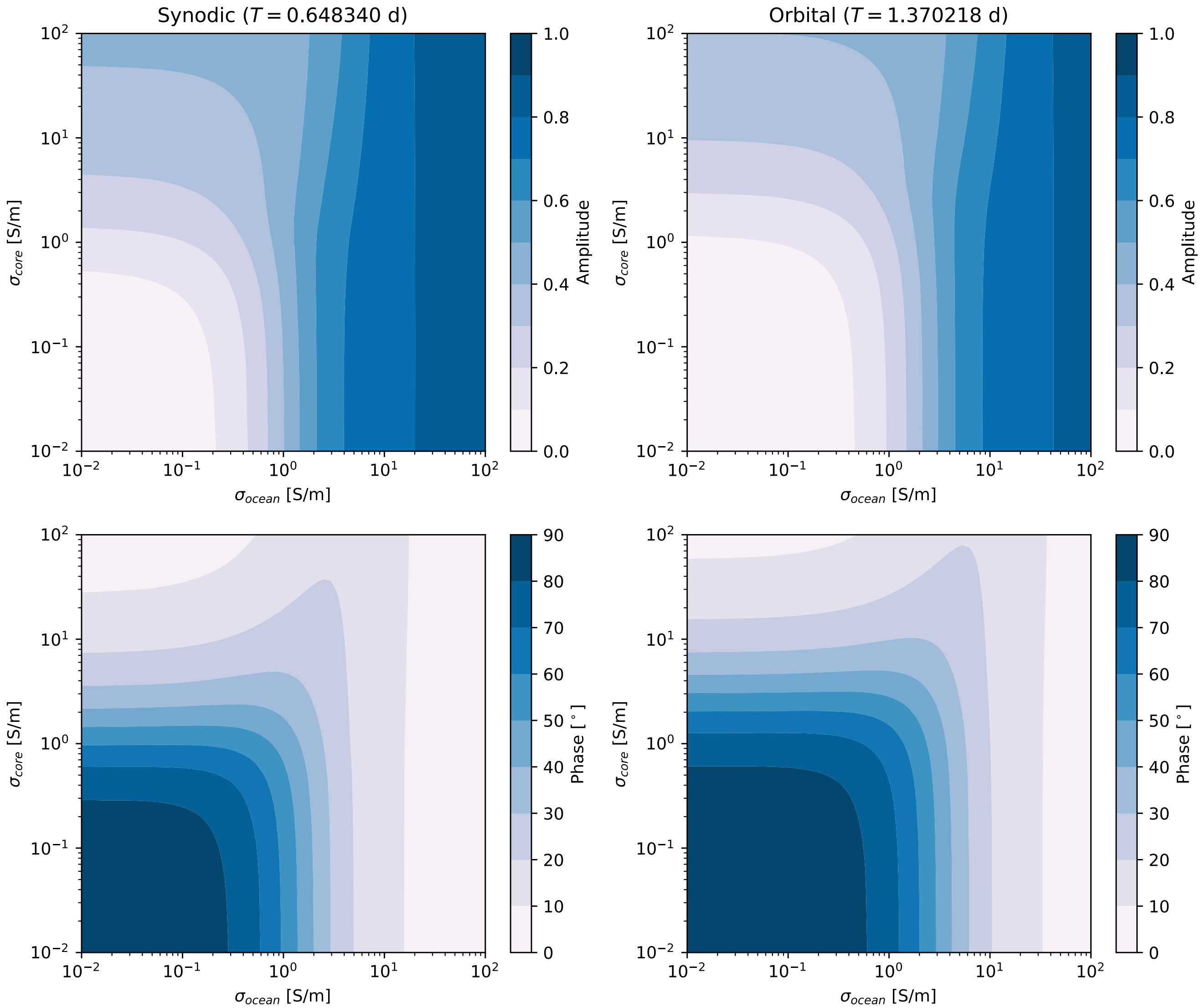

Figure 9. Induction amplitude and phase as a function of conductivity for a three-layer model consisting of a negligibly conducting crust (thickness 5 km), a conducting ocean (thickness 57 km), and a conducting core (radius 194 km). The conductivities of the core and the ocean are free parameters. The induction amplitude A and its phase are displayed for Saturn's synodic rotation and orbital periods as a function of ocean and core conductivities.

With increasing ocean conductivities, the effects of the core become less and less important and, starting around $3\,\mathrm{S\,m^{-1}}$, even have a small negative effect on the induction amplitude. For conductivities of $5\,\mathrm{S\,m^{-1}}$ of the ocean, which is roughly similar to the average conductivity of Earth's oceans of $3.2\,\mathrm{S\,m^{-1}}$, the induction amplitude is 0.55 and 0.47 for the synodic and orbital periods, respectively. The phase lags of the induced fields are significant with values of 20° and 31°, respectively.

5. Search for Induction Signatures during Close Enceladus Flybys

In this section we analyze the magnetic fields measured during Enceladus flybys of the Cassini spacecraft and search for induced signals from Enceladus's interior. We apply several of the time-variable inducing models derived and discussed in Section 3.

In our analysis we use two different coordinate systems. We use the Saturn body-fixed IAU system in spherical coordinates, i.e., radial distance r, colatitude Θ, and azimuth φ. We also use the Enceladus body-fixed IAU system (in Cartesian coordinates), as a body-fixed frame is appropriate for induction studies. However, we rotate the axis such that the new y-direction corresponds to the conventional IAU x-direction, i.e., roughly toward Saturn, and the new x-direction corresponds to the $-y$-direction of the IAU system, i.e., approximately along the orbital direction of Enceladus. The z-direction is in the direction of Saturn north. This system might be referred to as the "rotated Enceladus IAU system." The reason is that the rotated coordinates are similar to the Enceladus interaction system (ENIS). Past studies of Enceladus's plasma interaction and magnetic field environment have been generally performed in the ENIS coordinate system. In the ENIS the primary axis z is aligned with Saturn's spin axis, y is in the direction of the component of the Enceladus–Saturn vector perpendicular to z, and x completes the right-handed system, which is aligned with the rigid corotation direction. In the IAU system and the ENIS the axes are not exactly aligned owing to the orbital eccentricity of Enceladus. The maximum misalignment of the axes of both coordinate systems changes over decades and can be as large as

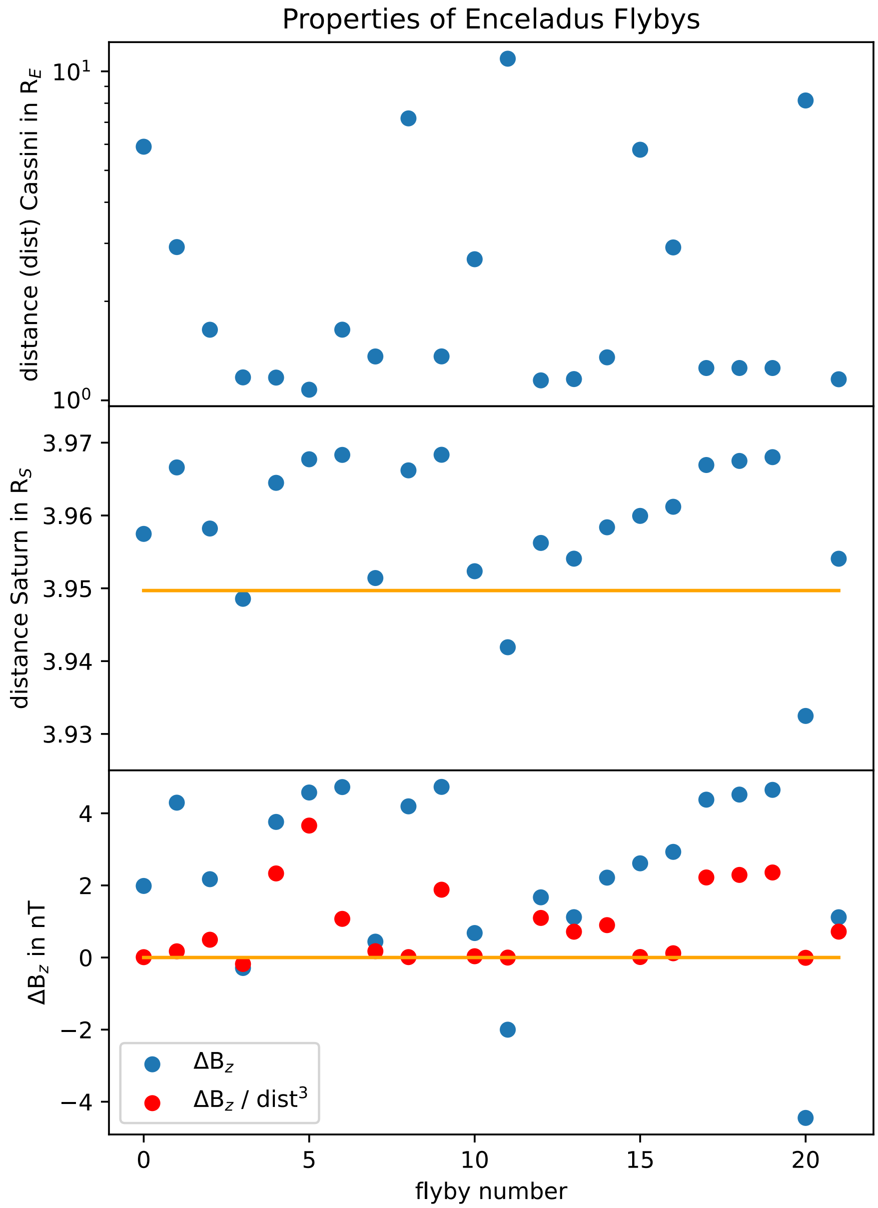

Figure 10. Overview of flyby properties as a function of flyby number. The top panel displays Cassini's distance from Enceladus at closest approach during flybys. The middle panel shows the distance of Enceladus from Saturn, with the orange line representing the average distance to Saturn. The bottom panel displays the expected amplitudes of the inducing field at the surface of Enceladus due to its eccentric orbit as blue circles. Red circles show an approximation for the expected induction amplitudes at closest approach taking into account the altitude of the flyby.

5°. Thus, the ENIS is not a body-fixed frame. In all figures where we display rotated Enceladus IAU coordinates, we label the axis also with standard Enceladus IAU coordinates to facilitate readability for both the planetary and space physics readership.

Locally homogeneous time-variable primary external fields induce dipole magnetic fields, which fall off with $(r/R_{\mathrm{E}})^{-3}$. Therefore, a necessary condition for the induced fields to be detectable is that the flyby occurred close to Enceladus. In Table 1 we refer to flybys within $1.3R_{\mathrm{E}}$ as close flybys. In Figure 10, we display the distance (dist) from the center of Enceladus at closest approach (C/A) for all flybys in the top panel. In the middle panel of Figure 10, we show the distance $r_{\mathrm{C/A}}$ of Enceladus from Saturn at C/A as a function of flyby number. The orange line shows Enceladus's average distance from Saturn r_{av}. The large majority of the flybys, except for three, occurred when Enceladus was at larger distances than r_{av}. This is due to a combination of the architecture of Cassini's orbit in the Saturn system and the precession period of Enceladus's pericenter with a period of 2.9 yr (R. Jacobson 2022).

In the bottom panel, we show the expected time-variable B_z component of the primary field due to Enceladus's eccentric orbit as blue circles. Because the amplitudes of the inducing field due to Enceladus's eccentric orbit are generally larger than the amplitudes from the PPO and LT variability, we illustratively show the effects of the eccentricity in Figure 10. Flybys with C/A at distances smaller than the average distance r_{av} lead to a negative perturbation, i.e., the $|B|$ field is larger and B_z more negative owing to the north–south orientation of Saturn's field. Flybys with distance larger than r_{av} experience a smaller field magnitude and thus a positive ΔB_z perturbation.

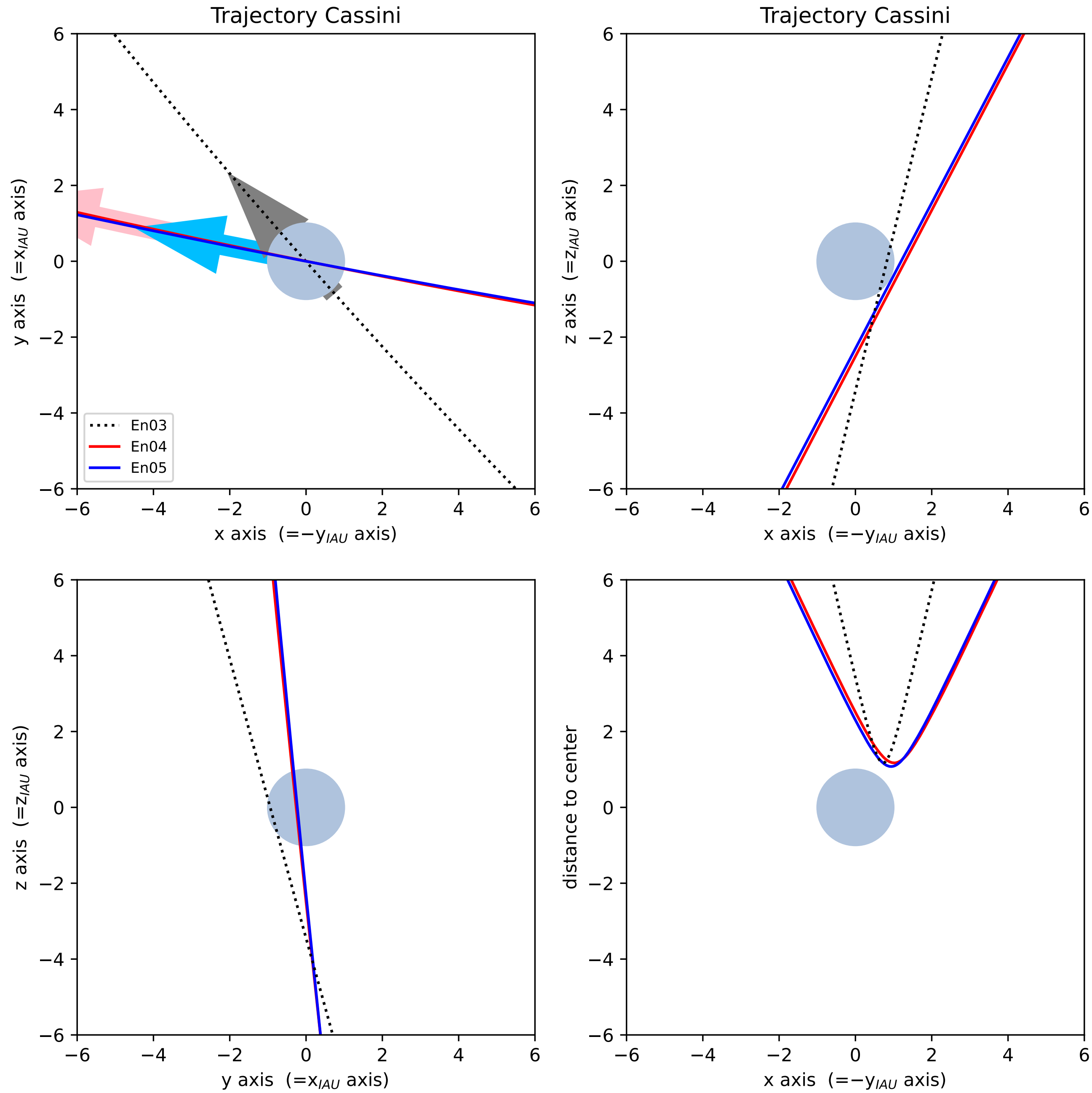

Figure 11. Trajectories of close north–south flybys En03, En04, and En05. The arrows indicate the direction of the Cassini spacecraft. Coordinates x, y, and z are in Enceladus IAU system rotated by 90° around the z-axis (definition see Section 5) with x approximately along Enceladus's orbital direction, y approximately towards Saturn, and z pointing north.

As induced fields fall off with the third power of the distance, we show the same ΔB_z perturbations weighted with $(R_{\mathrm{E}}/dist)^3$ as red circles. This demonstrates that only the very close flybys are suited to search for induction signals. It also shows that, based on the eccentric orbit, unfortunately no flybys are available where a negative ΔB_z acts as an inducing field. Thus, we cannot identify induction cases based on alternating directions of the induced fields, as was possible for Europa (M. G. Kivelson et al. 2000).

The flybys can also be categorized based on the trajectory of the flyby. We name these properties in Table 1. The close flybys relevant for the induction studies fall into the categories of south polar flybys, north polar flybys, and north–south flybys that cross both hemispheres of Enceladus. These three categories will be presented separately and explained in the following subsections.

5.1. North–South Flybys: En03, En04, En05

In this subsection, we present the three close flybys En03, En04, and En05. They have in common that their trajectories are directed in north–south direction as displayed in Figure 11. For the induced magnetic fields we assume in this subsection for simplicity an induction amplitude $A = 1$ and no phase lag. For the En04 and En05 flybys (Figures 12 and 13), the magnetic field measurements contain signatures that appear to be consistent with contributions from induction, while for the En03 flyby we only expect a very small induced signal based on our modeled primary inducing field (Figure 14). In all three

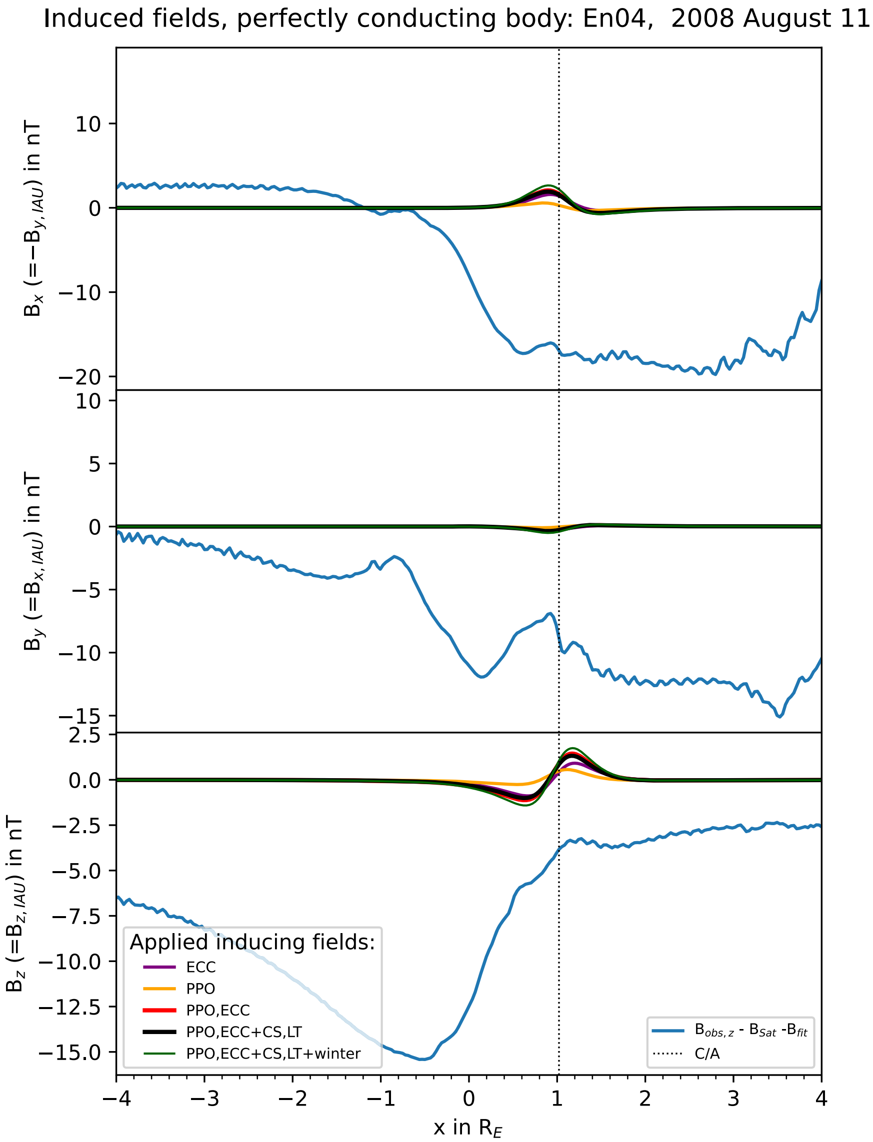

Figure 12. North–south flyby En04: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2. Near closest approach magnetic field perturbations are present, which could be consistent with modeled induced field.

figures, the magnetic field measurements are shown in rotated Enceladus body-fixed IAU Cartesian coordinates as blue lines. We subtracted Saturn's internal magnetic field calculated with Gauss coefficients derived in M. K. Dougherty et al. (2018). In the z-component of the magnetic field we additionally subtracted the contribution from Saturn's magnetodisk (see Section 3). Large magnetic field perturbations are visible in the observed fields. They primarily stem from the plasma interaction of Saturn's magnetospheric plasma with the plumes of Enceladus (e.g., M. K. Dougherty et al. 2005; K. K. Khurana et al. 2007; J. Saur et al. 2007; H. Kriegel et al. 2011; S. Simon et al. 2011). During close flybys, these perturbations are on the order of 10–30 nT. In Appendix B, we show results of magnetohydrodynamic (MHD) simulations under simplified geometry and conditions to help with the interpretation. These also show the large amplitudes of the magnetic field perturbations due to the interaction of Saturn's magnetospheric plasma with the plumes located near the south pole. All flybys shown in this work exhibit large negative B_x perturbations, which are typical for northern Alfvén wing passages (also see Appendix B). The reason for negative B_x measurements even during the south polar flybys is that in those cases Cassini was still mostly north of the center of the plumes and thus observed the northern Alfvén wings (e.g., M. K. Dougherty et al. 2006).

In all three flybys magnetic field perturbations on the scales of an Enceladus radius are present near closest approach (C/A). The closest approach is displayed as vertical dotted lines in the figures. Our model induced fields are displayed for the various assumptions about the primary inducing fields derived in Section 3. We show induced fields due to Enceladus's eccentric orbit and the PPO variability in purple and orange, respectively, as well as both combined in red. Assuming additionally current sheet effects (CS), which generally reduce the amplitudes due to eccentricity by 20%–30%, and including possible LT effects, the results are shown in black. If we further consider a seasonal dependence on top of the current sheet dependencies, the resulting induced response is shown as dark-green line. The black and dark-green curves are based on our

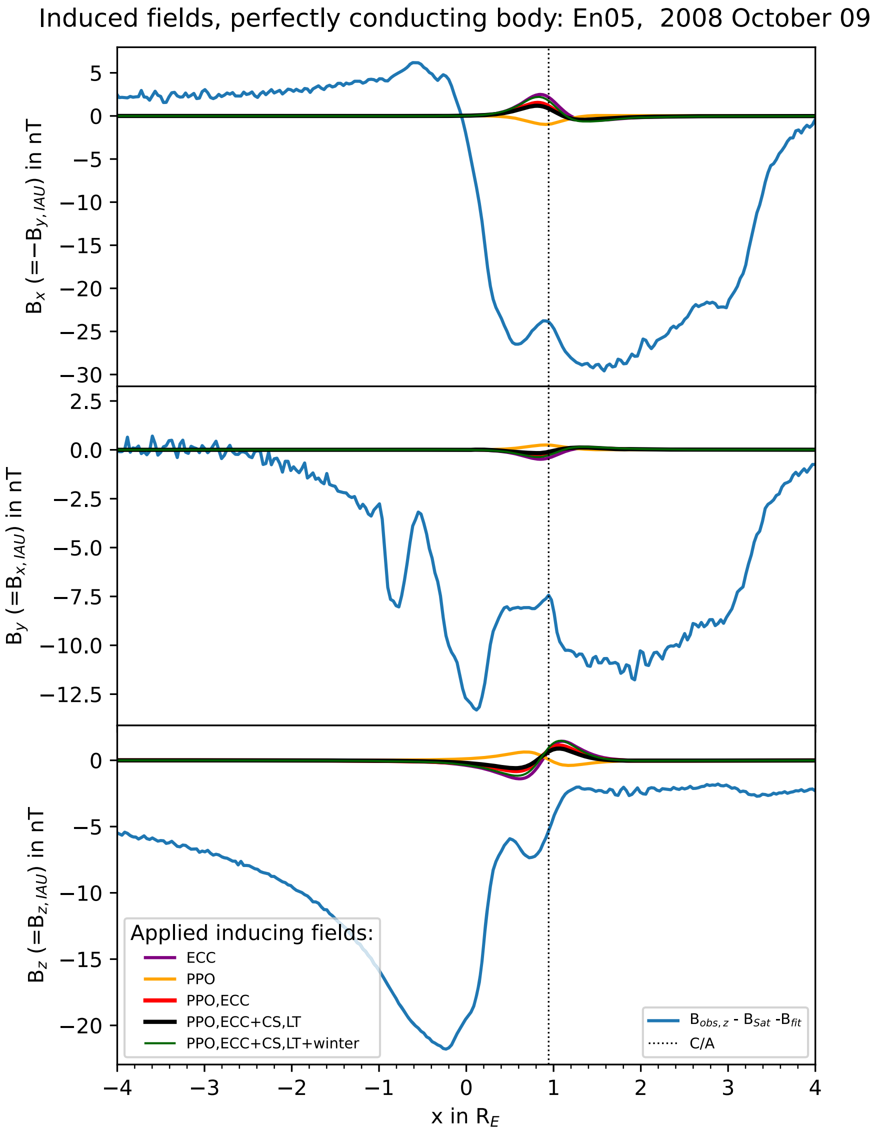

Figure 13. North–south flyby for En05: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2. Near closest approach magnetic field perturbations are present, which could be consistent with modeled induced field.

best/highest-level models of the inducing field and generally lead to induced fields more consistent with the data than other inducing models. Quantitative properties of the inducing field can be found in Tables 2 and 3. Based on the three figures, we see that the effects of individual primary field components are not always constructive, as their effects can partially cancel each other out.

The possible agreement between observed magnetic field perturbations and modeled induction fields is most prominent for the En05 flyby (Figure 13). This flyby shows a localized dip in the B_z component for $x < 1R_{\mathrm{E}}$. Similarly, there is an increase in the B_x component at a similar location also possibly consistent with an induced signal. While, to the authors' knowledge, no results of magnetic field simulations along the En03 and En04 flybys are available in the literature, H. Kriegel et al. (2011) show in their Figure 5(g) modeled fields along the En05 trajectory. In their modeled data no magnetic field perturbations at the location of the observed perturbations near C/A are visible. This indicates that plumes and plasma-related effects might not be the reason for the observed local perturbations around closest approach. However, we would like to add that the plasma interactions due to the various local plumes with different directions and time variability of their own are very difficult to fully predict, rendering an uncertainty for the interpretation of individual flybys. The En04 flyby occurred on a very similar trajectory to the En05 flyby. The observed magnetic field perturbations are somewhat weaker compared to En04. The modeled induced field has similar amplitudes for the En04 and En05 flybys; however, the magnetic field due to eccentricity, the most reliable of the predicted primary fields, is slightly weaker for En04 compared to En05.

The En03 flyby occurred along a somewhat different trajectory compared to En04 and En05. Therefore, the plasma magnetic fields are also expected to have a somewhat different structure. For the En03 flyby shown in Figure 13 we predict at best a very low amplitude induced magnetic field signature. While the B_x and B_y components of the observed field do not

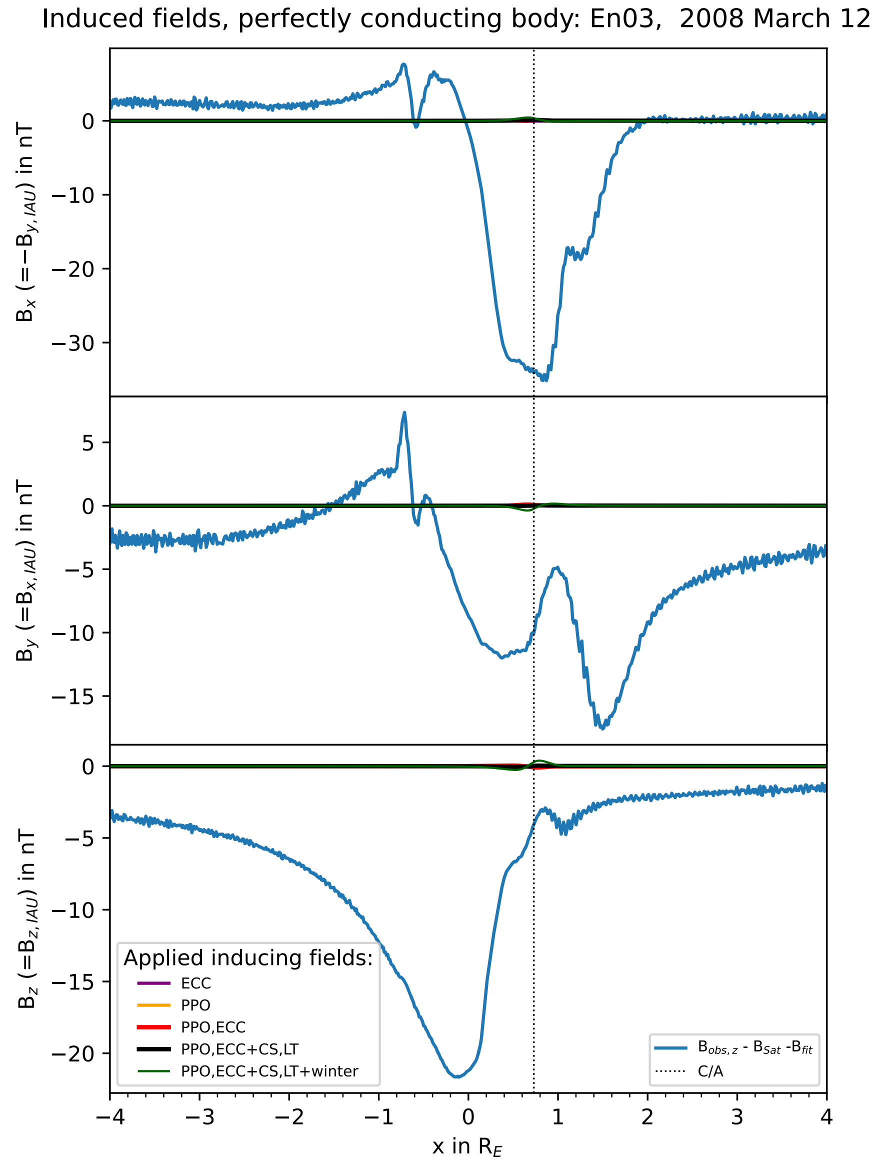

Figure 14. North–south flyby En03: magnetic field perturbations in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2.

show a particular feature near closest approach, the B_z component exhibits a local feature. In our interpretation, this feature is inconclusive; it might stem from induction due to an underpredicted primary field during the flyby, or it could be an effect of the plasma interaction with the plumes. The induced fields shown in this section for the perfectly conducting case are calculated based on the various models/assumptions for the primary inducing fields. The models have uncertainties, which are characterized by correlation coefficients and standard deviations shown in Table 2. The uncertainty in the inducing field generates an uncertainty in the inducing field, which we discuss quantitatively in Section 5.2.2.

5.2. North Polar Flybys: En12, En13

Cassini performed only two close flybys across Enceladus's north pole labeled En12 and En13. The trajectories are shown in Figure 15. These flybys are the best-suited ones for the search of induction signals because the plumes are located on the other hemisphere near the south pole. The magnetic field measurements for these two flybys we show as a function of y because the dominant direction of the Cassini spacecraft was in the direction away from Saturn.

5.2.1. Perfectly Inducing Case

The observed magnetic fields minus Saturn's internal magnetic field are displayed in Figures 16 and 17 for the En12 and En13 flybys, respectively. The main B_x and B_y perturbations have amplitudes of about -15 and -8 nT, respectively. These main perturbations stem from the plasma interaction with the plumes produced near the south pole, even though the solid body of Enceladus partially blocks the magnetic perturbations that propagate as MHD waves away from the plumes. The two key reasons for the existence of magnetic perturbations near the north pole are as follows: Some plume gas extends sufficiently far out that the resultant magnetic perturbations can pass Enceladus. Therefore, the

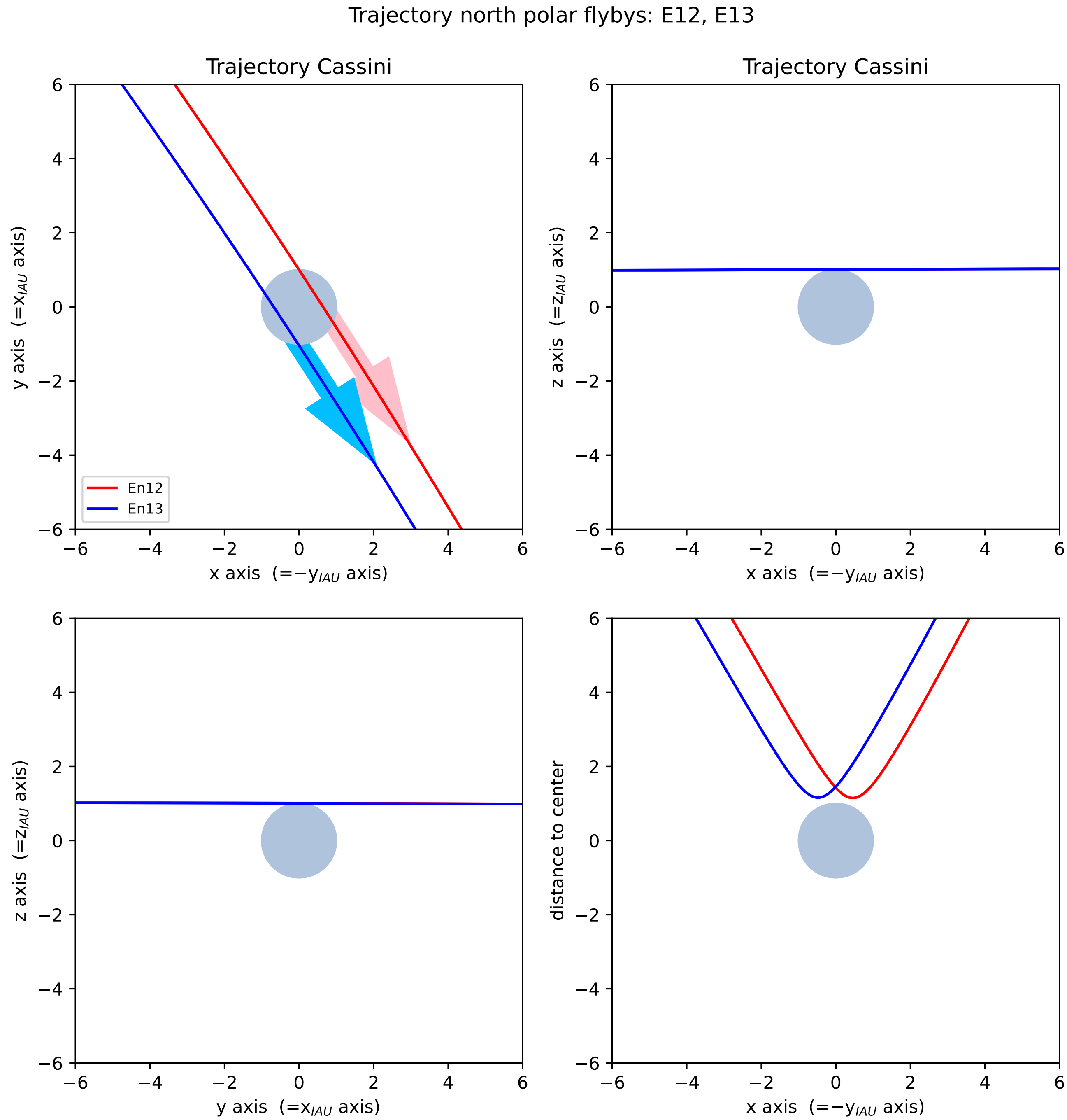

Figure 15. Trajectories of the close north polar flybys En12 and En13. The arrows indicate the direction of the Cassini spacecraft. Coordinates x, y, and z are in Enceladus IAU system rotated by 90° around the z-axis (definition see Section 5) with x approximately along Enceladus's orbital direction, y approximately towards Saturn, and z pointing north.

perturbations need to be generated outside of the tangent cylinder (or tube) given by the magnetic field lines tangent to Enceladus. For an unperturbed magnetic field assumed to be in the z-direction, the tangent cylinder (or tube) is given by $r_\perp^2 = x^2 + y^2 = R_\mathrm{E}^2$. Only outside of this tube can the plasma waves travel along Saturn's background magnetic field lines from the south to the north. The trajectory crossings through the tube boundaries are shown in Figures 16 and 17 as vertical gray lines. For example, for En12 the B_x and B_y perturbations on positions around and before $y = -1 R_\mathrm{E}$ are plasma effects generated in the south propagating along the magnetic field lines past Enceladus to the north. The second reason is the "hemispheric coupling effect" for magnetic field lines disconnected by a solid body (J. Saur et al. 2007). The most prominent examples are the large B_x and B_y components during En12 and E13 within the tube. They are Alfvénic perturbations associated with hemispheric coupling currents flowing tangent to Enceladus on the boundary of the tube (see Figure 4(b) in J. Saur et al. 2007). The sign of the B_y component reflects a reversed Hall effect due to negatively charged dust grains (H. Kriegel et al. 2011; S. Simon et al. 2011). The hemispheric coupling effects at Enceladus have been identified and analyzed in Cassini data (S. Simon et al. 2014). To the authors' knowledge, no dedicated simulation study explicitly showing

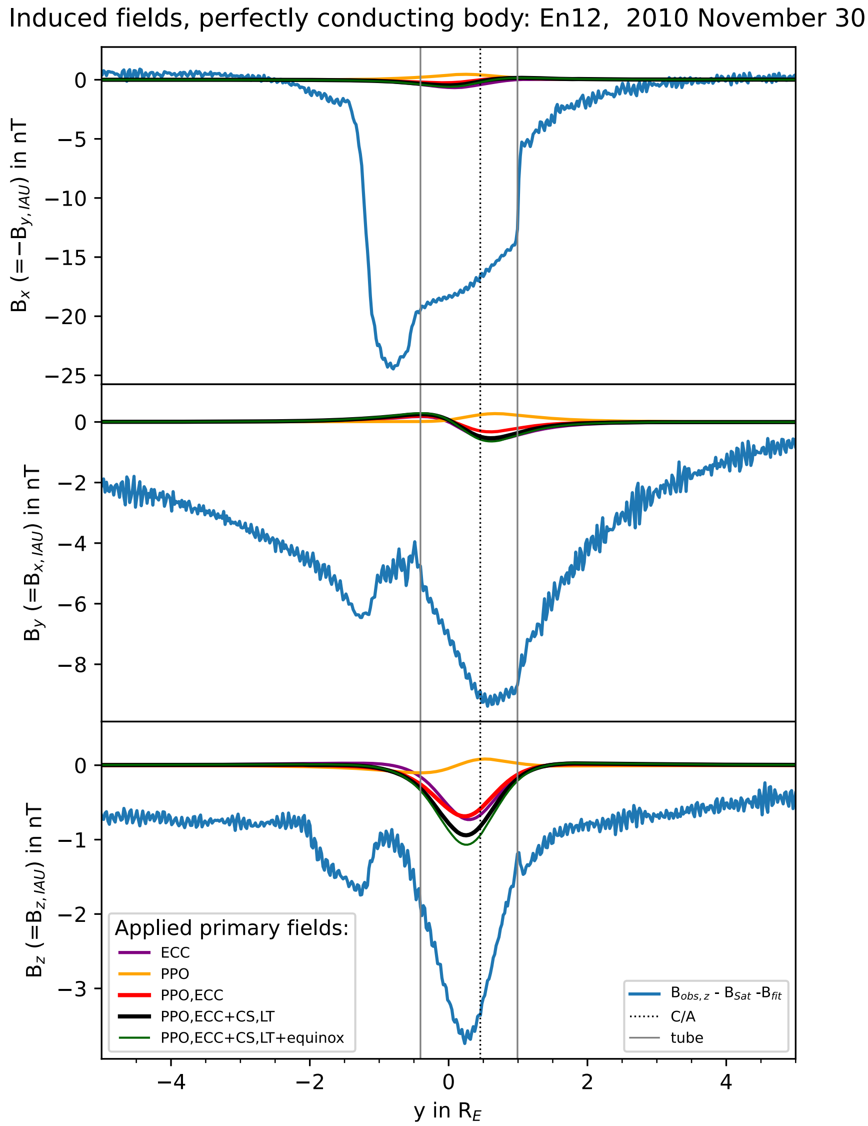

Figure 16. North polar flyby En12: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2. Near closest approach magnetic field perturbations are present, which could be consistent with modeled induced field.

magnetic field perturbations over the north pole of Enceladus is available in the literature. Therefore, we present in Appendix B idealized MHD simulations that confirm the picture of large B_x and B_y magnetic field fluctuation across the north pole despite the plumes being located at the south pole and Enceladus's solid body acting as a plasma wave absorber.

The B_z component for En12 and En13 is much less perturbed compared to the B_x and B_y components. Near the north pole the plasma perturbations from the plumes originating at the south pole are mostly Alfvénic in nature owing to the hemisphere coupling effects. Therefore, the magnetic perturbations within the tube are mostly incompressible, i.e., fluctuations of $|\boldsymbol{B}|$ and B_z caused by the plume plasma physics are very small. Thus, the B_z component over the north pole is more diagnostic than other components or data from other flyby locations. Magnetic field perturbations due to plasma effects still arise as a result of pressure balance effects that occur within tubes that are cooled or depleted of plasma. The idealized MHD simulations shown in Appendix B demonstrate this effect as well, which we will further quantify in Section 5.2.2.

The B_x and B_y components for the En12 and En13 flybys shown in Figures 16 and 17 contain substructures that could be consistent with those caused by induction. Even though the inducing, primary field is predominantly parallel to the z-direction, the induced fields have a dipolar structure also leading to field components in the x- and y-directions. For the B_z component during the En12 flyby (Figure 16), the modeled perturbation due to eccentricity within Saturn's unperturbed internal field (purple) is in the direction of the observed perturbations, while those from the PPO fields are mildly in antiphase (orange). The combined induced field due to eccentricity and PPO fields is shown in red and contributes significantly to the observed B_z component. When considering current sheet and LT effects, the joint induced field (black) becomes stronger. Considering seasonal effects in constraining the contribution from the current sheet and the periodic LT

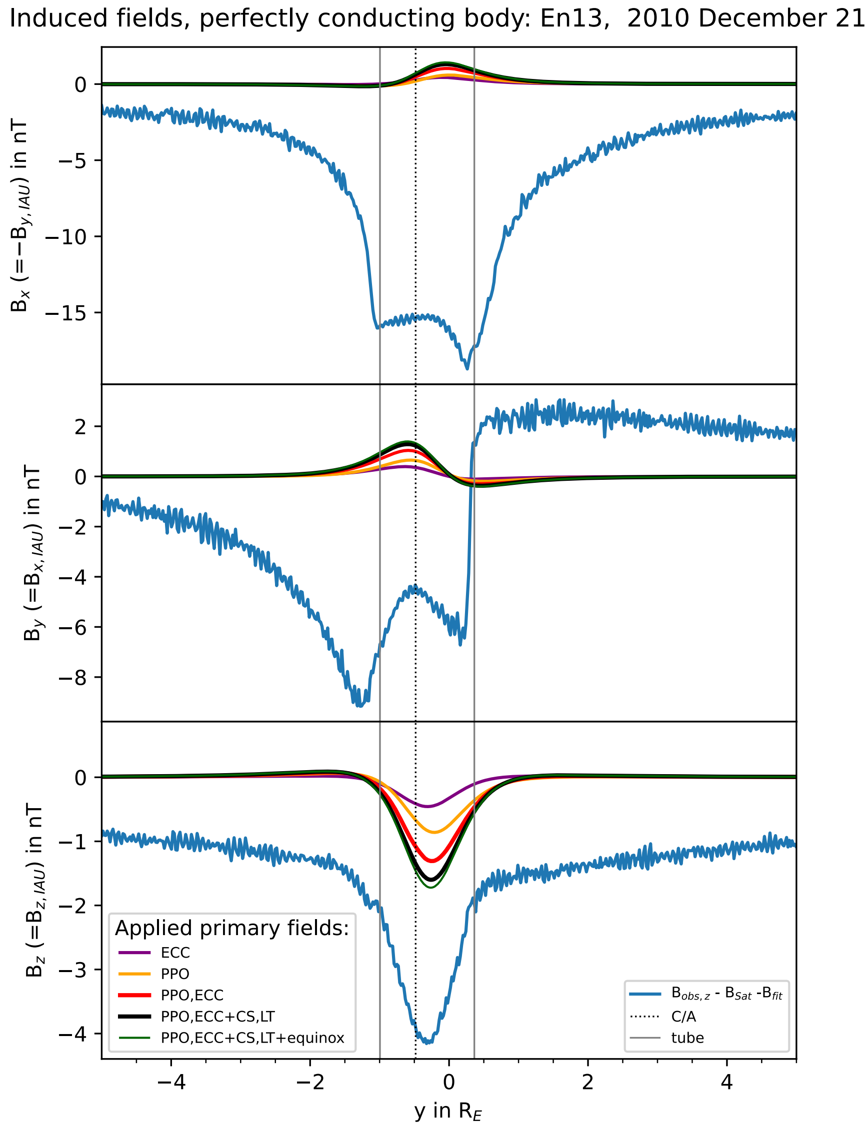

Figure 17. North polar flyby En13: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2. Near closest approach magnetic field perturbations are present, which could be consistent with modeled induced field.

effects leads to the dark-green curve, i.e., the strongest induced field.

As shown in Figure 17, for the En13 flyby the PPO fields (orange) generate induced fields in the direction of the observed perturbations. Eccentricity effects (purple) are also in the same direction. The induced field from PPO and eccentricity (red) covers a significant part of the observed field in the B_z component. Considering an improved primary model with current sheet and its LT variability (black) leads to induction fields closer to the observed field and even slightly improved fits when specifying these fields to the equinox epoch (in dark green).

5.2.2. Finite Conductivity

Up to now we simply assumed Enceladus as a perfectly conducting body to search for induction signals. Now we investigate the implications of finite conductivities inside of Enceladus and discuss the effects of the plasma pressure that contribute to the magnetic field perturbations in the B_z component.

In Figures 18 and 19, we only show all induced field contributions combined but calculated for various internal conductivities with the three-layer model introduced in Section 4. We use the primary inducing field model displayed as crosses in Figure 8 and as black curves in Figures 16 and 17 for a perfectly conductive body. This inducing model includes PPO and eccentricity effects within the current sheet and LT effects. We show induced fields for an ocean and a core with various conductivity combinations. In the labels of Figures 18 and 19 showing these conductivities, the first conductivity value is for the ocean and the second one (after “&”) is the conductivity value for the core. We also show the infinite conductive case as black lines. We shift the background offset for a better comparison. For En12 and En13 flybys, the conductivities of 1 S m^{-1} for the ocean and the core lead to negligible induced field signals not consistent with the observations. For conductivities on the order of 1 S m^{-1}, and due to the small size of Enceladus, large phase lags ranging

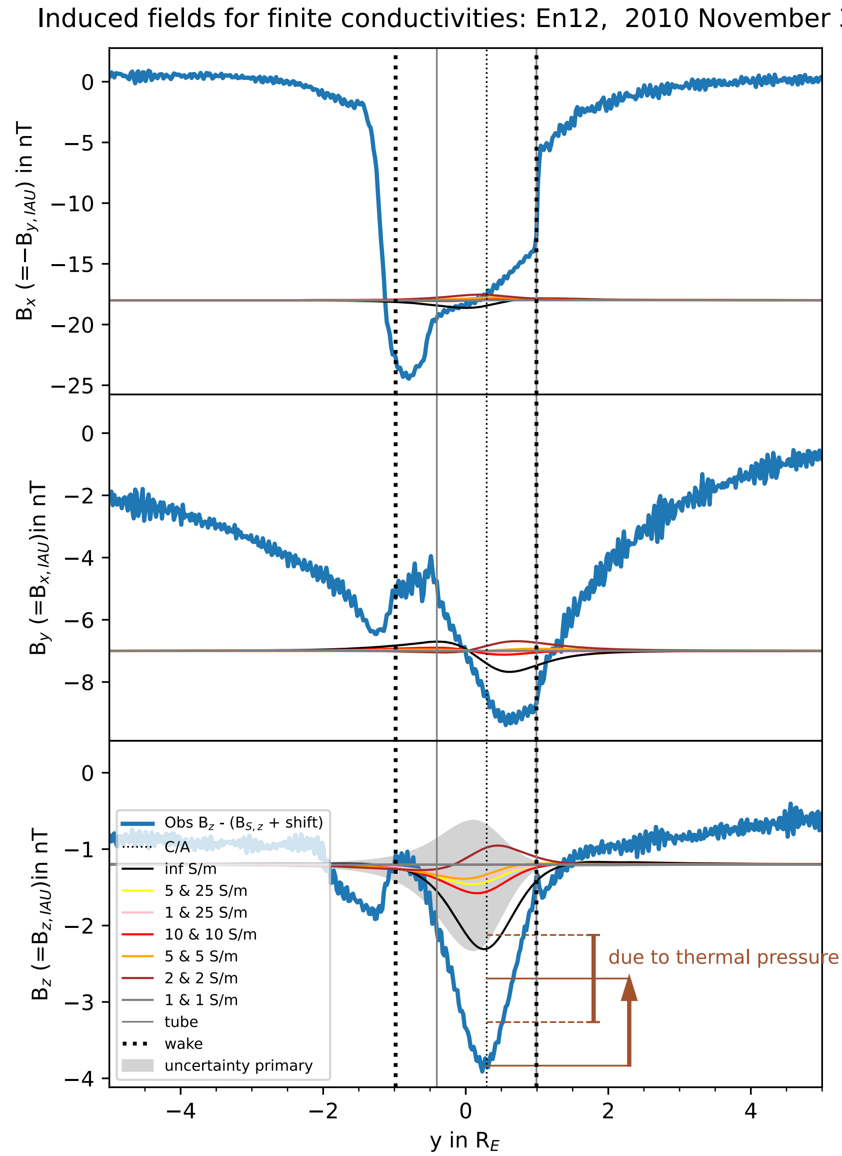

Figure 18. Finite conductivity study for flyby En12. In the labels displaying conductivities, the first number is the conductivity of the ocean and the second number (after "&") is the conductivity of the core. The observed B_z component in blue is artificially shifted to directly cross-compare the local observed magnetic field perturbations with the modeled induced fields. The vertical olive-brown arrow displays the contribution due to thermal pressure, and the horizontal dashed lines represent its associated uncertainties. The gray area shows exemplarily the uncertainty range around the 5 & 25 S m^{-1} case. The sum of our modeled induced fields (gray region) and the magnetic pressure (brown arrow) including their uncertainties is sufficient to account for the observed perturbations shown in blue.

from roughly 30° to 70° for the synodic and orbital periods occur (see Figure 9). These phase lags additionally reduce the amplitude of the induction response. For similar conductivities, the phase lags play a larger role at Enceladus owing to its smaller size compared to those occurring at the larger Galilean satellites (C. Zimmer et al. 2000). Also for the En12 flyby, 2 S m^{-1} for the ocean and the core does not generate a clear induction signature. Either 10 S m^{-1} for the ocean and core or a 1 S m^{-1} ocean conductivity and a highly conductive core of 25 S m^{-1} generate an induction signal of about 0.5 nT amplitude for the En12 flyby and about 1 nT for the En13 flyby. The predicted induction fields in both cases are not large enough to fully explain the observed B_z perturbations. The missing contributions could come from magnetic pressure perturbations and/or an underprediction of the inducing primary field. We will discuss both in the following paragraphs.

Within the tube of field lines threading Enceladus, plasma moves along the magnetic field and is partially absorbed by Enceladus. This leads to a partially empty tube that can extend further downstream as a partially empty wake. Wake boundaries defined by $|y| < 1R_{\mathrm{E}}$ are shown as dashed lines and tube boundaries as gray lines in Figures 18 and 19. During the En13 flyby, no wake was crossed. An empty or partially depleted tube or wake would generate magnetic field perturbations to compensate for the lost plasma pressure by magnetic pressure as seen at other moons (e.g., J. Saur et al. 2002; S. Simon et al. 2012; K. K. Khurana et al. 2017). The thermal pressure of the ions during the En12 and En13 flybys is about $(3 \pm 1.5) \times 10^{-10}$ Pa based on Cassini CAPS data.[5] The pressure of the energetic

[5] The Cassini CAPS data used were from the Cassini-Huygens Cassini Plasma Spectrometer (CAPS) Derived Ion Moments Data Collection (J. H. Waite & J. Furman 2022).

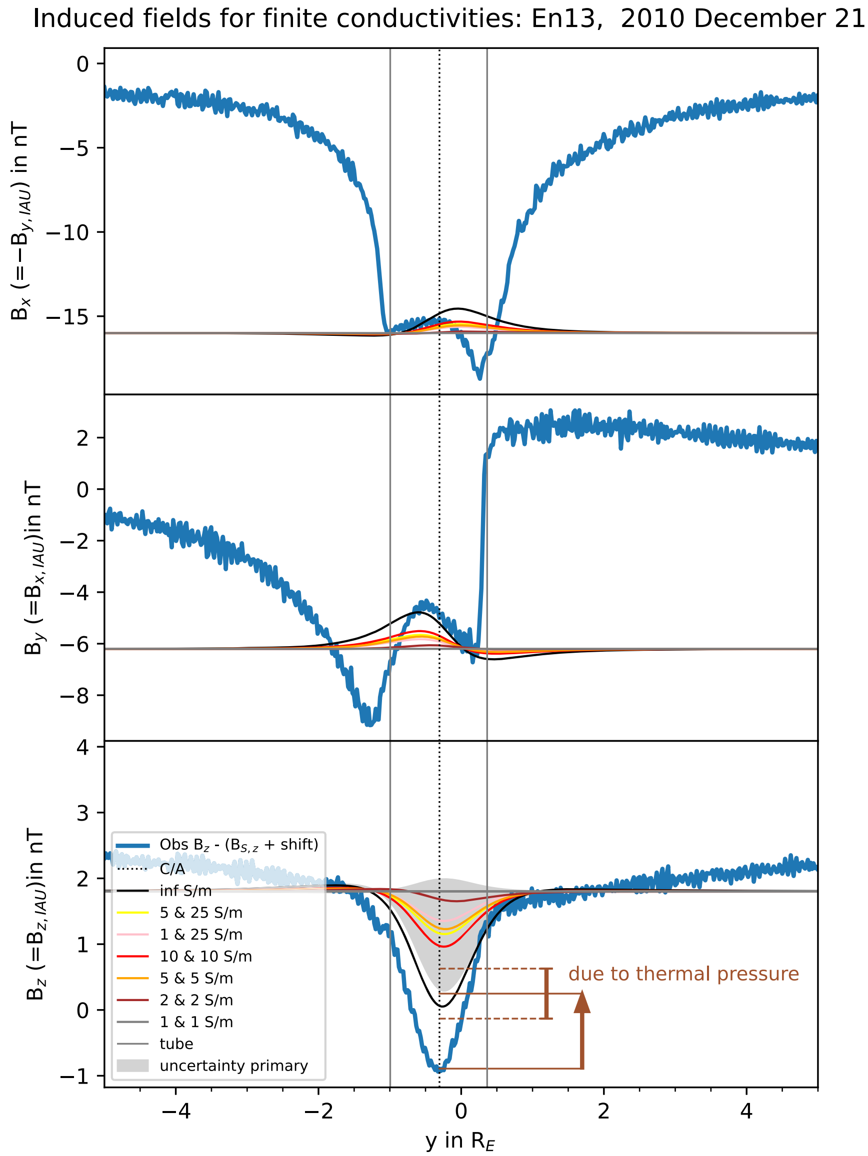

Figure 19. Finite conductivity study for flyby En13. In the labels displaying conductivities, the first number is the conductivity of the ocean and the second number (after "&") is the conductivity of the core. The observed B_z component in blue is artificially shifted to directly cross-compare the local observed magnetic field perturbations with the modeled induced fields. The vertical olive-brown arrow displays the contribution due to thermal pressure, and the horizontal dashed lines represent its associated uncertainties. The gray area shows exemplarily the uncertainty range around the 5 & 25 S m^{-1} case. The sum of our modeled induced fields (gray region) and the magnetic pressure (brown arrow) including their uncertainties is sufficient to account for the observed perturbations shown in blue.

particles in the keV range (N. Sergis et al. 2007) is negligible compared to the thermal pressure. Pressure balance requires $\nabla((B_0 + \delta B)^2/2\mu_0 + (p_0 + \delta p)) = 0$. As an upper limit, we can assume that the flux tube is completely emptied such that the pressure perturbation is $\delta p = -p_0$, with p_0 the pressure outside the tube. Neglecting terms quadratic in δB, the resultant magnetic field perturbations are $\delta B = -\mu_0 \delta p/B_0$. With a background field of about 330 nT, the magnetic pressure perturbations would be $\delta B = -1.1 \pm 0.6$ nT. They are aligned with the z-direction because B_0 is in the z-direction. In Figures 18 and 19, we show δB resulting from the magnetic pressure as the olive-brown arrow. The upper and lower dashed vertical lines represent the uncertainty range of ± 0.6 nT.

The magnetic pressure effects are calculated as upper limits, i.e., the plasma tube/wake is only locally and completely emptied. Including the magnetic pressure effect, the observations in the z-direction could be consistent with induction within a perfectly or very highly conductive body, but it would still require conductivity values larger than the values of the finite conductivity scenarios of a few S m^{-1} for the ocean and up to 25 S m^{-1} for the core presented in Figures 18 and 19. The discrepancy between predicted induced fields within finite conductive layers and observed fields minus possible plasma pressure effects within its error bars is about 0.5 nT for the En12 flyby and about 0.3 nT for the En13 flyby.

The inducing primary fields in the case of Enceladus might be stronger than derived with our model. The orbital eccentricity effects are deterministic to a high degree since the orbit of Enceladus and the internal field are well-known. Our applied inducing model, shown as crosses in Figure 8, is based on variability due to PPO, eccentricity, and LT effects in the current sheet, but it still includes significant scatter with a standard deviation $c_\sigma = 2.4$ nT for the B_z component (see Table 2). They might be due to solar wind effects, mass loading

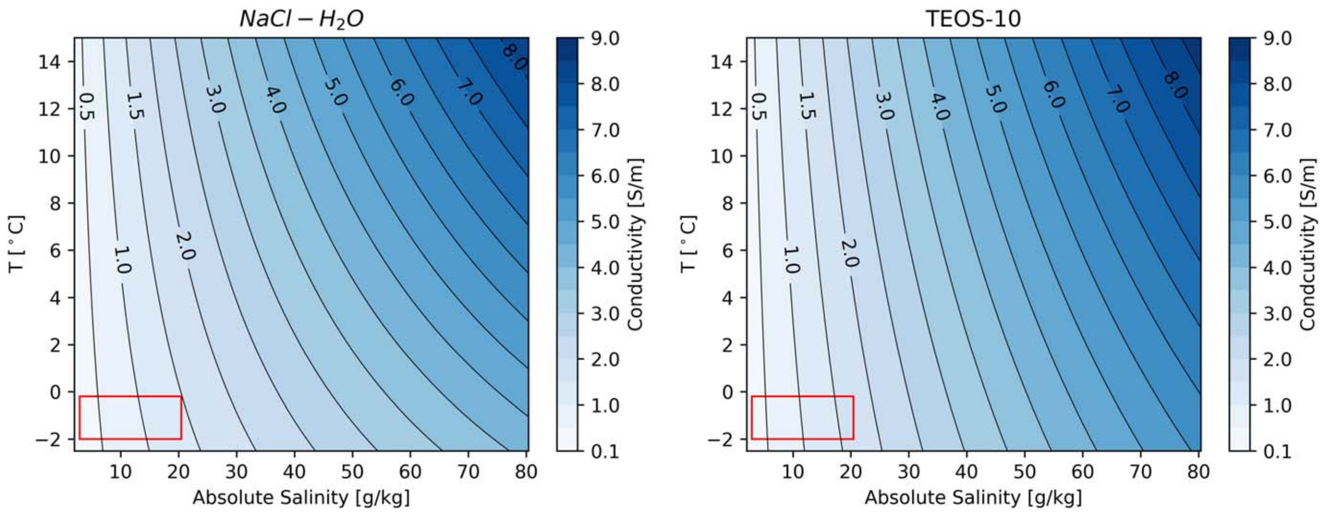

Figure 20. Conductivity range of Enceladus's ocean. Both panels show conductivities as a function of salinity and temperature for an ocean assumed to contain only NaCl (left) or a composition similar to seawater on Earth, i.e., based on the thermodynamic equation of state of seawater TEOS-10 (right). The red boxes show the parameter range expected based on current observations and geophysical understanding of Enceladus (more see text).

effects in the Enceladus torus, or dynamic plasma transport effects in the magnetosphere, leading to an incomplete description of the periodic amplitudes and phases of our primary field models. In addition, the PPO fields, which represent well-established fits to the data, exhibit significant deviations from the PPO model in the form of scatter (e.g., D. J. Andrews et al. 2019). In order to estimate the effects of the uncertainty in the induced fields, we use the standard deviation c_σ in the B_z (or B_Θ) component and assume that it represents inducing field. This is likely an overestimation since a fraction of the standard deviation c_σ could be a time-variable component on timescales longer than the orbital period and the associated magnetic field would not or less contribute to induction. Assuming now that $c_\sigma = 2.4$ nT represents inducing uncertainty, we propagate this uncertainty within an interior model of 5 S m^{-1} for the ocean and a highly conductive core of 25 S m^{-1} into the induced fields and display the propagated uncertainties along the En12 and En13 flybys on top of the expected signal for the same interior conductivity model. The resultant uncertainty is shown as the gray area around the yellow lines in Figures 18 and 19. Including the uncertainty of the primary inducing field, we find induction responses for the En12 and En13 flybys that are consistent within the error bars of the pressure-corrected observations, i.e., the gray shaded region overlaps with the upper error bar from magnetic pressure shown as the dashed olive-brown line. It is important to stress that if no induction occurs and the amplitude $A = 0$, then the uncertainty area completely disappears as well. The reason is that any primary field and any uncertainty of it still generate no induced field for $A = 0$. In this case, the magnetic pressure effect alone is not able to explain the observed magnetic field perturbations in the B_z component, supporting a contribution from induction.

The MHD simulations shown in Appendix B also support our interpretation of the south polar flybys. Figure 22 displays in the last panel that the plasma thermal pressure p is reduced over the north pole. The magnetic field perturbations in the third panel show a decrease in the B_z component of about -0.5 nT for a run with a background pressure $p_0 = 0.3$ nPa (dark solid line, default run labeled Setup 1). The magnitude of the magnetic field shown in the fourth panel shows the overall increase of the magnetic pressure to compensate for the decrease in the plasma pressure. In the simplified MHD simulations of Setup 1, the magnetic field perturbations are about 50% of those from the upper limit of a tube with fully reduced pressure. However, this number is not universal as shown by several other model runs displayed and discussed in Appendix B. In Setups 3a and 3b we change the upstream plasma pressure p_0, which leads to smaller pressure perturbations and thus also to smaller perturbations of the B_z component. These runs demonstrate that a reduction of plasma thermal pressure is compensated by an associated smaller increase in magnetic pressure. Generally, the plasma pressure perturbations depend on how plasma is cooled in its interaction with the surface and also on how strongly the plasma is slowed owing to the plumes in the south pole. Electron pressure effects will increase the magnetic pressure effects to some degree as well. The main purpose of the idealized simulations is to show that across Enceladus's north pole the plasma interaction generates only small magnetic perturbations in B_z due to pressure balance effects, which are not large enough to explain the observed B_z perturbations. Thus, other processes must occur in addition. Induction within the interior of Enceladus is consistent with being the main missing component.

With the simulations in Appendix B we also investigate the role of the plumes in shaping the magnetic field environment over the north polar region. In simulation Setups 1, 2a, 2b, 2c, and 2d we apply various plume densities ranging from $n_{n,0} = 0.3 \times 10^9$ cm^{-3} to 30×10^9 cm^{-3} while keeping all other simulation parameters constant. Our standard plume model with a plume surface density of $n_{n,0} = 3 \times 10^9$ cm^{-3} is shown as solid black lines (Setup 1) in Figure 22. The runs of Setups 2a, 2b, 2c, and 2d are displayed as purple lines. These numerical experiments demonstrate that in the northern Alfvén wing the plume density still significantly controls the B_x and B_y components. Larger plume densities lead to larger amplitudes of the B_x and B_y perturbations. The B_x component can have amplitudes as large as around -30 nT over the north pole. In contrast, the B_z component is only very mildly affected by the plumes. An increase in plume density does not increase the amplitude of a negative B_z perturbation but reduces the perturbation and can even change its sign for Setup 2d with

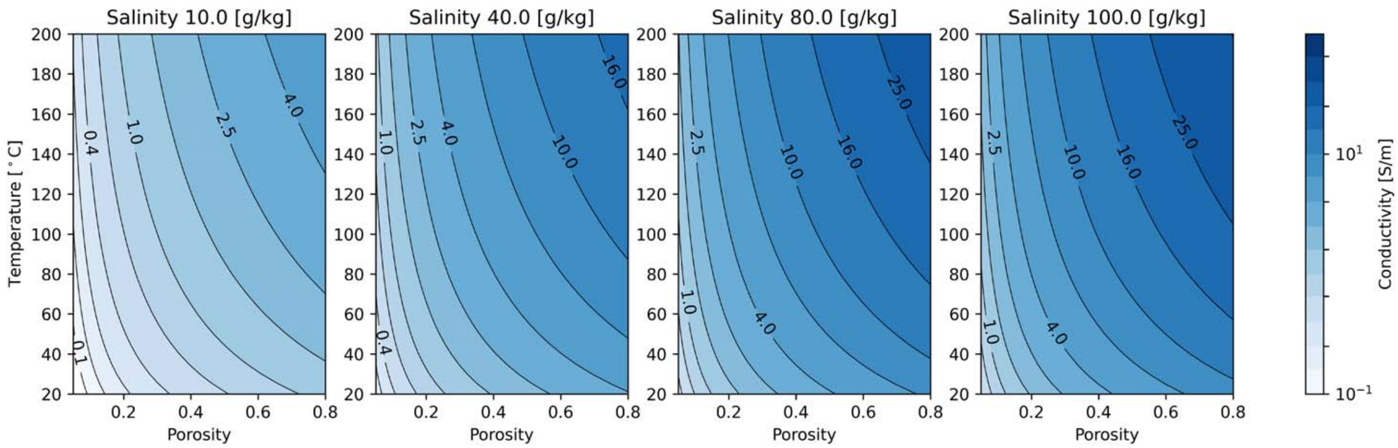

Figure 21. Bulk conductivity range of Enceladus's core as a function of its porosity and temperature for various assumptions about its salinity. The two-phase system fluid is NaCl–H_2O.

the largest plume density. The reason is that the Alfvénic interaction causes a small positive B_z perturbation in the north polar Alfvén wing (e.g., J. Saur et al. 2007), which adds to the negative B_z component owing to the pressure balances. Overall the simulations shown in Appendix B demonstrate that only very small negative perturbations in the B_z component with amplitudes up to -0.7 nT are generated by the MHD interaction. Thus, the best location and the best-suited field component to monitor induction effects from Enceladus is the B_z component over the north polar region.

6. South Polar Flybys: En14, En17, En18, En19

The south polar flybys are dominated by the plasma interaction within the complex systems of plumes (J. Spitale & C. Porco 2007). These flybys are thus the most difficult ones to identify induction signals. The trajectories of these flybys are displayed in Appendix C in Figure 23, showing that En14, En17, and En18 have very similar trajectories. The observed magnetic fields during these flybys and the induction models are shown in Appendix C in Figures 24–27. Around closest approach there is a positive peak in the B_z component for the En14, En17, and En18 flybys. In contrast, the induced fields show a negative peak around closest approach. However, magnetic perturbations along the En14 and En17 flybys calculated with a hybrid model by H. Kriegel et al. (2014) show larger plasma peaks near C/A compared to the observed peaks. Thus, a contribution from induction in the negative z-direction would be consistent with the observations. A similar consistency argument can be given for the En19 flyby, for which H. Kriegel et al. (2014) model weaker magnetic field perturbations near closest approach compared to the observations. Thus, an additional contribution from induction would bring the total modeled field closer to the observations. These flybys could be consistent with induction signals but are overall inconclusive owing to the large role of the plume physics in shaping the magnetic fields on Cassini's trajectories through the plumes. Therefore, we put no further effort into analyzing these flybys.

7. Implication for Enceladus's Interior

The magnetic field observations during the flybys studied here indicate that the induction response of Enceladus needs to be rather strong to fit the observations. After consideration for the in situ plasma magnetic field effects, the preferred scenario for the electromagnetic induction response is a high amplitude and a small phase. From Figure 9, we see that this combination is achieved either when the ocean layer has a high electrical conductivity, rendering conductivity of the core less important owing to fast attenuation of fields, or for an ocean with a moderate conductivity with a highly conductive core underneath. In this section, we discuss both of these scenarios in the context of additional observations.

The electrical conductivity of the ocean depends on salinity and temperature (pressure in the ocean affects the conductivity very little within the considered range). Following evolution and thermodynamic models, the temperature of Enceladus's ocean is assumed to lie between -2°C and 1°C (C. R. Glein et al. 2018). Observational constraints on concentration and composition of salts are derived primarily from the Cassini observations over the south polar plumes (F. Postberg et al. 2011), although it is debated whether these values are fully representative of the actual salinities in the ocean. Some studies suggest that the actual range can be broader and vary owing to ocean dynamics and stratification (W. Kang et al. 2022b). In this work, we will use two models to derive the conductivity of the ocean: (i) the thermodynamic equation of state of seawater TEOS-10 (R. Feistel 2018), and (ii) a model of a NaCl–H_2O fluid. The major identified salt compounds in the plume are NaCl, $NaHCO_3$–Na_2CO_3, and KCl (F. Postberg et al. 2011), with salinity ranges given by C. R. Glein et al. (2018). For relevant temperatures and salinities, the electrical conductivity of all these salt solutions is rather similar, within a factor of two (R. B. McCleskey 2011). Therefore, we calculate the conductivity of the mixture of identified salts using recalculated equivalent NaCl concentrations and a laboratory model of P. N. Sen & P. A. Goode (1992), specifically their Equation (9). This empirical relationship was found to be suitable for temperatures and salinity values considered here (N. Watanabe et al. 2021). Alternatively, we can use the international thermodynamic equation of seawater TEOS-10 (Intergovernmental Oceanographic Commission 2011; R. H. Tyler et al. 2017; R. Feistel 2018; A. V. Grayver 2021) to model Enceladus's ocean conductivity. Although there is no reason to assume that the actual composition of Enceladus's ocean is similar to the seawater, it is instructive to use it as a reference owing to its high quality. Figure 20 shows electrical conductivity over relevant temperature/salinity ranges.

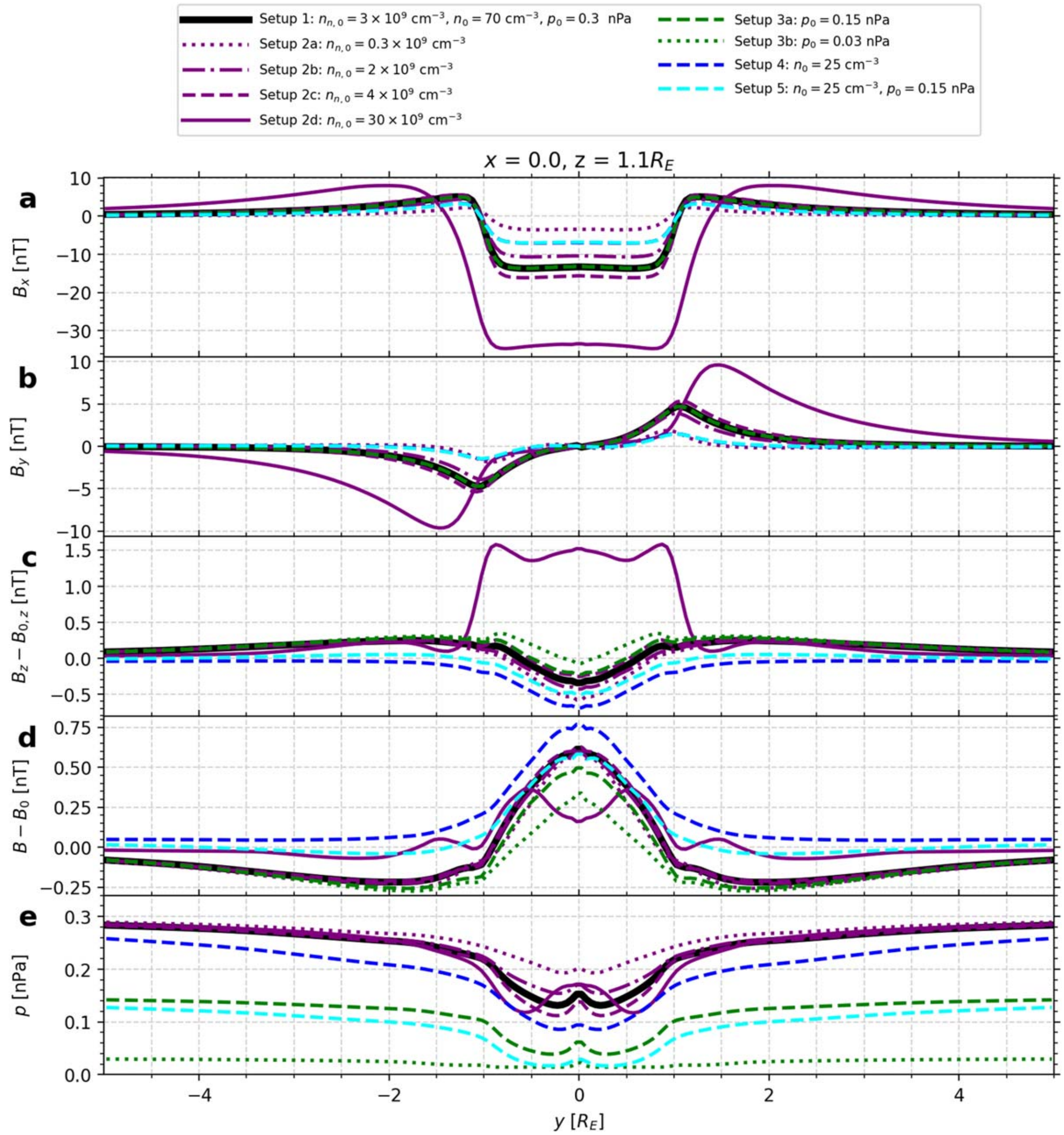

Figure 22. Simulated magnetic field perturbations over Enceladus's north polar region obtained with idealized MHD modeling. Simulation setups are summarized in Table B1.

The conductivity of the core is less constrained and depends not only on the properties of the fluid phase but also on the porosity and interconnectedness within the pore space. A high porosity is required to sustain the hydrothermal circulation in the interior of Enceladus (G. Choblet et al. 2017). These authors assumed a 30% uniform porosity throughout the core. Later studies by I. Kisvárdai et al. (2023) and W. Neumann & A. Kruse (2019) proposed a radially dependent porosity with a higher porosity near the ocean−core boundary. Both scenarios can explain the Cassini gravity and orbital dynamics observations. The conductivity of the rock phase (chondritic/silicate composition) is likely much lower (G. Parthasarathy & S. Sharma 2004) than that of the fluid phase unless there exists a hydrothermal alternation resulting in the formation of electrically conductive alteration products (e.g., clays). Electrical conduction in the presence of fluids and clays becomes rather complex (Y. Qi & Y. Wu 2022), and in view of a lack of real constraints on these processes in Enceladus, we will assume that the conductivity of the solid phase is 1×10^{-3} S m^{-1}. Due to a high porosity and hypothesized hydrothermal circulation within the core, we assume that pores are interconnected. Therefore, we calculate the bulk conductivity of a solid−fluid system using the upper Hashin–Strikmann bound (Z. Hashin & S. Shtrikman 1962). Figure 21 shows the bulk conductivity of the two-phase system for a range of porosities and temperatures that are typically assumed for the Enceladus core (G. Choblet et al. 2017; W. Neumann & A. Kruse 2019; I. Kisvárdai et al. 2023). Figure 20 shows that with current observations on the plume salinity (F. Postberg et al. 2011; C. R. Glein et al. 2018), and assuming that this range is relevant for the ocean, the conductivity of the ocean is <2 S m^{-1} for a plausible temperature range, for both the NaCl–H_2O solution and seawater. The associated range of conductivities is displayed within the red boxes in Figure 20. These conductivities would not suffice to produce a sufficiently strong induction response to explain field perturbations observed near C/A during the flybys discussed here. With a moderately conductive ocean, time-varying magnetic fields will also penetrate to the core and induce secondary fields there. The core of Enceladus is believed to be hydrothermally active (G. Choblet et al. 2017), which requires high porosity and temperature. In turn, this implies that the bulk core conductivity may reach high values, potentially much higher than that of the ocean owing to higher temperatures and salinities. This is illustrated in Figure 21, where higher temperatures and salinities may lead to bulk conductivities of up to 20–30 S m^{-1} for the core. This, however, requires higher salinity values than those inferred for the plume. As was pointed out by W. Kang et al. (2022b), the salinity of Enceladus's ocean may not be fully constrained by the plume values, and the actual values can be higher. This also holds for the core, where not only the concentration of detected salts can be higher but also other conductivity-enhancing solubles are potentially present (J. C. Castillo-Rogez et al. 2022).

The results of our analysis of the north polar flybys in Section 5.2.2 suggest two scenarios of Enceladus's interior. If the in situ salt measurements within the plumes represent the salinity of Enceladus's ocean, then a highly conductive core of 25 S m^{-1} or more is necessary to generate a sufficient induction response. This would represent a geophysically plausible scenario of Enceladus's interior as discussed in the previous paragraph. Alternatively, the salinity of the ocean might be larger than derived from the plume measurements. However, for the expected temperature of the ocean of around and below 0°C and for plausible salinities, the conductivity cannot reach values much in excess of 5 S m^{-1}.

The values for the conductivities of the ocean and the core were derived based on the comparison of measured and modeled B_z components over the north polar region. In Figures 18 and 19, the modeled induced fields, combined with the fields resulting from pressure balance, including their error bars, overlap with measured magnetic field perturbations. The assumptions in calculating the induced fields, those from pressure balance, and their associated uncertainties are discussed at length in Section 5.2.2. However, these uncertainties need to be kept in mind, and measurements by a future mission to Enceladus are necessary to confirm with certainty the induced signal and to improve the conductivity models of Enceladus's interior.

8. Summary and Conclusions

In our work we present the first dedicated search for electromagnetic induction signals at Enceladus. We analyze magnetic field measurements obtained during Enceladus's flybys and orbit crossings by the Cassini spacecraft and demonstrate that despite the azimuthal symmetry of Saturn's internal magnetic field, several time-periodic fields can be identified in the magnetic field measurements. These are (i) the PPOs occurring on the synodic rotation period of Saturn, (ii) effects due to Enceladus's eccentric orbit, and (iii) LT asymmetries within Saturn's magnetospheric plasma sheet, with the latter two occurring on the orbital period of Enceladus. After identification of these inducing fields and their uncertainties, we calculate expected induction responses within a perfectly conductive Enceladus and within realistic conductivities of Enceladus's ocean and its core. We find within magnetic field measurements obtained during close flybys by the Cassini spacecraft perturbations which appear to be consistent with such induction responses. As the plasma interaction with Enceladus's plumes produces magnetic field perturbations about one order of magnitude larger compared to those expected from induction, an uncertainty in the identification of induced signals in particular on Enceladus's southern hemisphere exists. For the north polar flybys the MHD effects of the external plasma are not sufficient to explain the observed magnetic field perturbations in the B_z component. Our modeled induced magnetic fields from within Enceladus are consistent with being the main missing contributions to explain the observed B_z perturbations of the north polar flybys. Contributions from non-MHD plasma effects, however, deserve attention in future studies, which is beyond of the scope of this work. The amplitudes of the possible induction signals identified in the magnetic observations suggest that Enceladus's porous core is highly conductive on the order of 20–30 S m^{-1}, as an ocean based on the salinity derived from the in situ plume measurements would not suffice to generate necessary induction responses. Alternatively, the salinity and conductivity of the ocean might be larger than derived from the plume measurements.

Our study also provides detailed strategies for future measurements to further help probe the habitability of Enceladus (such as suggested in, e.g., M. L. Cable et al. 2021; G. Choblet et al. 2022; A. H. Sulaiman et al. 2022; Z. Martins et al. 2024). The exploration of Enceladus's interior with magnetic sounding can be executed best by well-timed close flybys across Enceladus's north pole or ideally with a polar orbiter. Such an orbiter could monitor the plume activity in the south polar region and induction in the north polar region. The north polar region is best "shielded" from the large plasma magnetic fields generated by the interaction with the plumes near the south pole. The solid body of Enceladus blocks the MHD waves generated near the south pole. Due to the hemispheric coupling effects (J. Saur et al. 2007; S. Simon et al. 2014), primarily the noncompressive MHD modes are carried into the north hemisphere, generating magnetic field perturbations perpendicular to Saturn's background magnetic field. Thus, magnetic field perturbations parallel to Saturn's background magnetic field are well suited to probe for induction signatures. In addition to the suggested magnetic field observations, plasma density and temperature measurements to constrain the buildup of magnetic pressure perturbations are important. Finally, flybys at various mean anomalies and LTs during the same seasonal epoch of Saturn will enable magnetic sounding from primary inducing fields with alternating directions to help better separate induction from non-induction-related contributions of Enceladus's magnetic field environment. For example, observations of inducing fields at various phases and alternating directions have convincingly established induction within Europa as the origin of its associated dipole perturbations (M. G. Kivelson et al. 2000). The strategy outlined here will thus conclusively allow us to identify and characterize induced fields from Enceladus's interior as an alternative means to probe a planetary body deemed to possess all necessary ingredients for habitability.

Acknowledgments

J.S. and S.D. received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No. 884711). A.G. was supported by the Heisenberg Grant from the German Research Foundation, Deutsche Forschungsgemeinschaft (project No. 465486300) and ESA Swarm DISC project No. 4000109587.

Appendix A
Radial and Latitudinal Dependence of Magnetic Field near Enceladus's Orbit

The sensitivity of magnetic field variability on the location of measurements near the orbit of Enceladus can be studied by Taylor expansion of Saturn´s internal field. Saturn's magnetic field can be written (following M. K. Dougherty et al. 2018) as

$$B_r(r,\Theta) = \sum_{n=1}^{n_{\max}} (n+1)\left(\frac{R_S}{r}\right)^{n+2} g_n^0\, P_n^0(\cos\Theta) \tag{A1}$$

$$B_\Theta(r,\Theta) = -\sum_{n=1}^{n_{\max}} \left(\frac{R_S}{r}\right)^{n+2} g_n^0\, \frac{\partial P_n^0(\cos\Theta)}{\partial\Theta} \tag{A2}$$

$$B_\phi(r,\Theta) = 0, \tag{A3}$$

where we used that Saturn's field is azimuthally symmetric within the measurement uncertainties and only coefficients of order zero are finite, i.e., $g_n^0 \neq 0$ and thus only associated Legendre polynomials P_n^0 of order zero are needed to describe the magnetic field (M. K. Dougherty et al. 2018). The coefficients have been determined up to degree $n_{\max} = 12$ (M. K. Dougherty et al. 2018). Now we investigate the variability $\delta\boldsymbol{B}$ defined through $\boldsymbol{B}(r_{\rm av}+\delta r,\ \Theta_{\rm av}+\delta\Theta) = \boldsymbol{B}(r_{\rm av}, \Theta_{\rm av}) + \delta\boldsymbol{B}$ by Taylor-expanding the field around the average position of Enceladus $r = r_{\rm av}$ and $\Theta = \Theta_{\rm av} = \pi/2$. Using Equations (A1)–(A3), we find for the perturbations $\delta\boldsymbol{B}$ of the magnetic field around the average position of Enceladus on its orbit around Saturn

$$\delta B_r(\delta r,\delta\Theta)|_{r_{\rm av},\Theta_{\rm av}} = \sum_{n=1}^{n_{\max}} (n+1)\left(\frac{R_S}{r_{\rm av}}\right)^{n+2} g_n^0 \times \left[(-n-2)P_n^0(\cos\Theta_{\rm av})\frac{\delta r}{R_S} + \left.\frac{\partial P_n^0(\cos\Theta)}{\partial\Theta}\right|_{\Theta_{\rm av}} \delta\Theta\right] \tag{A4}$$

$$\delta B_\Theta(\delta r,\delta\Theta)|_{r_{\rm av},\Theta_{\rm av}} = -\sum_{n=1}^{n_{\max}} \left(\frac{R_S}{r_{\rm av}}\right)^{n+2} g_n^0 \times \left[(-n-2)\left.\frac{\partial P_n^0(\cos\Theta)}{\partial\Theta}\right|_{\Theta_{\rm av}} \frac{\delta r}{R_S} + \left.\frac{\partial^2 P_n^0(\cos\Theta)}{\partial\Theta^2}\right|_{\Theta_{\rm av}} \delta\Theta\right] \tag{A5}$$

$$\delta B_\phi(\delta r,\delta\Theta)|_{r_{\rm av},\Theta_{\rm av}} = 0. \tag{A6}$$

In the set of Saturn's Gaussian coefficients g_n^0, the coefficient g_1^0 is about an order of magnitude larger than any other coefficients with degree $n > 1$ (M. K. Dougherty et al. 2018). Additionally, fields associated with larger degrees n have falloffs owing to the dependence of $\left(\frac{R_S}{r_{\rm av}}\right)^{n+2}$ in the above expansions. Therefore, we simplify Equations (A4) and (A5) using only $n = 1$ terms to

$$\delta B_r(\delta r,\delta\Theta)|_{r_{\rm av},\pi/2} = -2\left(\frac{R_S}{r_{\rm av}}\right)^3 g_1^0\ \delta\Theta \tag{A7}$$

$$\delta B_\Theta(\delta r,\delta\Theta)|_{r_{\rm av},\pi/2} = -3\left(\frac{R_S}{r_{\rm av}}\right)^3 g_1^0\ \frac{\delta r}{R_S}, \tag{A8}$$

where we have used that the inclination of Enceladus is $i = 0.0°$ (R. Jacobson 2022), i.e., $\Theta_{\rm av} = \pi/2$. Equation (A7) shows that the variability in B_r in Figure 2 is due not to the eccentricity of Enceladus but largely to crossings of the Cassini spacecraft above or below Enceladus's spin plane. On an ellipse with semimajor axis a, the maximum and minimum distances from its foci are $a(1+e)$ and $a(1-e)$, respectively. Equation (A8) thus can be used to approximate the maximum and minimum variability Enceladus is exposed to as

$$\delta B_\Theta = \pm 3(R_S/R_{\rm av})^3 g_1^0\ e \approx \pm 3B_0\ e, \tag{A9}$$

with B_0 the average azimuthal field along its orbit.

Due to current sheet effects, mostly the latitudinal field component has contributions in addition to Saturn's internal field. If the resultant field can locally be described by a falloff with a power law of the form $B_\Theta = B_0(R_s/r)^\beta$ with exponent β, then, similarly to Equations (A8) and (A9), this field can be Taylor-expanded along the orbit of Enceladus as $\delta B_\Theta \approx \pm\,\beta B_0\,\delta r/R_S$, and measurements of δB_Θ can be used as an estimate of the power-law exponent β.

Appendix B
Idealized MHD Simulations over Enceladus's North Pole

To further support why the B_z component over the north pole is well suited to search for induction signals and to help the reader get an impression of the magnetic field perturbations generated by the plasma interaction with Enceladus's plumes, we performed simplified MHD simulations. In Appendix B.1 we briefly introduce our model and its setups. In Appendix B.2 we discuss the effects of variable plume densities and the role of pressure balance on the magnetic field over the north polar region.

B.1. MHD Model Setup

For the simulation an MHD model with a number of simplifications was applied. The model is similar to standard moon–magnetosphere interaction models (e.g., X. Jia et al. 2008; S. Duling et al. 2014, 2022) and solves the single-fluid MHD equations without the Hall effect. We turned off any internal field. The boundary conditions at Enceladus's surface are absorbing and insulating, i.e., no electric current can penetrate the insulating icy crust. We used a simplified neutral gas model to represent the effects of the plumes near the south pole. The neutral densities n_n of the plumes are described by an azimuthally symmetric model with $n_n(r,\Theta_s) = n_{n,0}\,(R_{\rm E}/r)^2 \times \exp(-(\Theta_s/H_{\Theta_s})^2) \times \exp(-(r-R_{\rm E})/H_d)$ and $\Theta_s = (180° -$ colatitude on Enceladus), i.e., $\Theta_s = 0°$ at the south pole and $\Theta_s = 90°$ at the equator. We choose $H_\Theta = 12°$, $H_d = 2R_{\rm E}$, and various values for the density $n_{n,0}$ in the plume center at the surface of Enceladus with typical values for $n_{n,0}$ discussed in the literature (e.g., J. Saur et al. 2008). They are listed in Table B1. The other parameters for the MHD simulations were chosen with the following values: the

Table B1
Values for Different Setups of MHD Simulations

Name	Properties	n_0 (cm^{-3})	p_0 (nPa)	$n_{n,0}$ (10^9 cm^{-3})
Setup 1	default	70	0.3	3.0
Setup 2a	plume	70	0.3	0.3
Setup 2b	plume	70	0.3	2.0
Setup 2c	plume	70	0.3	4.0
Setup 2d	plume	70	0.3	30.0
Setup 3a	pressure	70	0.15	3.0
Setup 3b	pressure	70	0.03	3.0
Setup 4	plasma density	25	0.3	3.0
Setup 5	plasma density and pressure	25	0.15	3.0

Note. The upstream plasma density is denoted n_0, its pressure is denoted p_0, and the density of the neutrals in the center of plume at the surface of Enceladus is denoted $n_{n,0}$ (for more details, see text).

upstream background magnetic field strength has only a vertical component of $B_{0,z}$ = –340 nT, the upstream plasma is characterized by an ion density $n_0 = 70\ cm^{-3}$, an average ion mass of 17.6 amu, and an upstream plasma velocity $v_0 = 26.4\ km\ s^{-1}$. We varied the plasma pressure p_0 (see various setups in Table B1). We used an ionization frequency $\nu_{ion} = 1.6 \times 10^{-8}\ s^{-1}$ of H_2O neutrals with molecular mass $m_{ion} = 16$ amu, an ion–neutral collision rate $\rho\, v\, \nu_{col}$ with frequency $\nu_{col} = k\, n_n$ with $k = 1 \times 10^{-8}\ cm^3\ s^{-1}$, and recombination with a rate coefficient $\alpha = 1 \times 10^{-13}\ m^3\ s^{-1}$.

B.2. Results of MHD Simulations

In Figure 22, we show modeled magnetic field perturbations and plasma pressure along the y-axis over the north polar region at $z = 1.1 R_E$ for a number of setups. The parameters used in these setups are summarized in Table B1. Simulation Setup 1 is our default setup.

We begin with investigating the role of the south polar plumes on the north polar magnetic field environment. Therefore, we perform simulations with a set of different plume densities ranging from $n_{n,0} = 0.3 \times 10^9\ cm^{-3}$ to $30 \times 10^9\ cm^{-3}$ in Setups 2a, 2b, 2c, and 2d while keeping all other parameters constant (see Table B1). The default run (Setup 1) representing typical values for Enceladus is shown as black solid lines in Figure 22. The run shows B_x and B_y values typical of a northern Alfvén wing, with B_x assuming values around −10 nT. The B_z perturbations are very small and decrease from about 0.2 nT around $|y| = 1 R_E$ to −0.3 nT. Setups 2a, 2b, 2c, and 2d are shown in purple. These model runs clearly demonstrate that significant plasma magnetic field perturbations in the B_x and B_y components exist over the north polar region for all model setups even though the plumes are located on the other hemisphere. The reason is that Alfvénic perturbations are transmitted through hemispheric coupling effects around Enceladus's absorbing and insulating body to the north polar region (J. Saur et al. 2007; S. Simon et al. 2014). Our set of runs 2a, 2b, 2c, and 2d also demonstrates that the Alfvénic interaction seen in B_x and B_y increases in amplitude with growing plume densities. However, the B_z perturbation has the largest negative amplitude for the smallest plume density. The B_z perturbation changes sign for the largest neutral density of Setup 2d with a value of 1.5 nT (and not consistent with the observed fields during En12 and En13). The reason is that the Alfvénic interaction causes a small positive B_z perturbation (see, e.g., Figure 4(b) in J. Saur et al. 2007). The larger the plume densities, the larger the Alfvénic interaction that contributes positive B_z perturbations. But most importantly for all plume densities, the B_z perturbations are small up to a local decrease of around −0.6 nT. Thus, the observed B_z perturbations of around −3 nT cannot be explained by plume variability.

Now we investigate the role of the upstream magnetospheric plasma while keeping the plume density constant. In Setups 3a and 3b we reduce the upstream plasma pressure p_0 compared to our default run (Setup 1). The results are shown as dark-green lines in Figure 22. We see that the plasma pressure does not affect the Alfvénic interaction, i.e., the B_x and B_y components are nearly identical. However, the magnitude of the B_z component is affected by the thermal plasma pressure p shown in the last panel of Figure 22. The plasma pressure is decreased over the north polar region due to absorption and cooling. (The small bump around $y = 0$ is due to a numerical resolution effect along the polar axis of the applied MHD code in spherical coordinates.) Runs of Setups 3a and 3b result in smaller plasma pressure perturbations and generate smaller B_z perturbations (in dark green) compared to the default run. This dependence confirms that locally reduced thermal plasma pressure is compensated by enhanced magnetic pressure, leading to larger magnetic field magnitudes and thus negative B_z perturbations. Next, we reduce the upstream density and keep all other parameters similar compared to the default run of Setup 1. The plasma density reduces the Alfvénic interaction and leads to B_z perturbations of around −0.7 nT (dashed blue lines in Figure 22). Finally, we show with Setup 5 (as dashed light-blue lines) results of a reduced upstream density and pressure. It shows slightly smaller B_z perturbations compared to Setup 4, confirming again the pressure balance effect.

With these simulations, we demonstrate that the plumes have no significant impact on the B_z component over the north pole. The B_z component is mostly controlled by pressure balance effects. Thus, the B_z component over the north polar region is well suited to search for low-amplitude signatures from induction generated within Enceladus.

Appendix C
South Polar Flybys: Observed Magnetic Fields and Modeled Induced Fields

The relevant properties and data of the close south polar flybys En14, En17, En18, and En19 are shown in this appendix. These include their trajectories in Figure 23 and the observed fields and modeled induced fields in Figures 24–27.

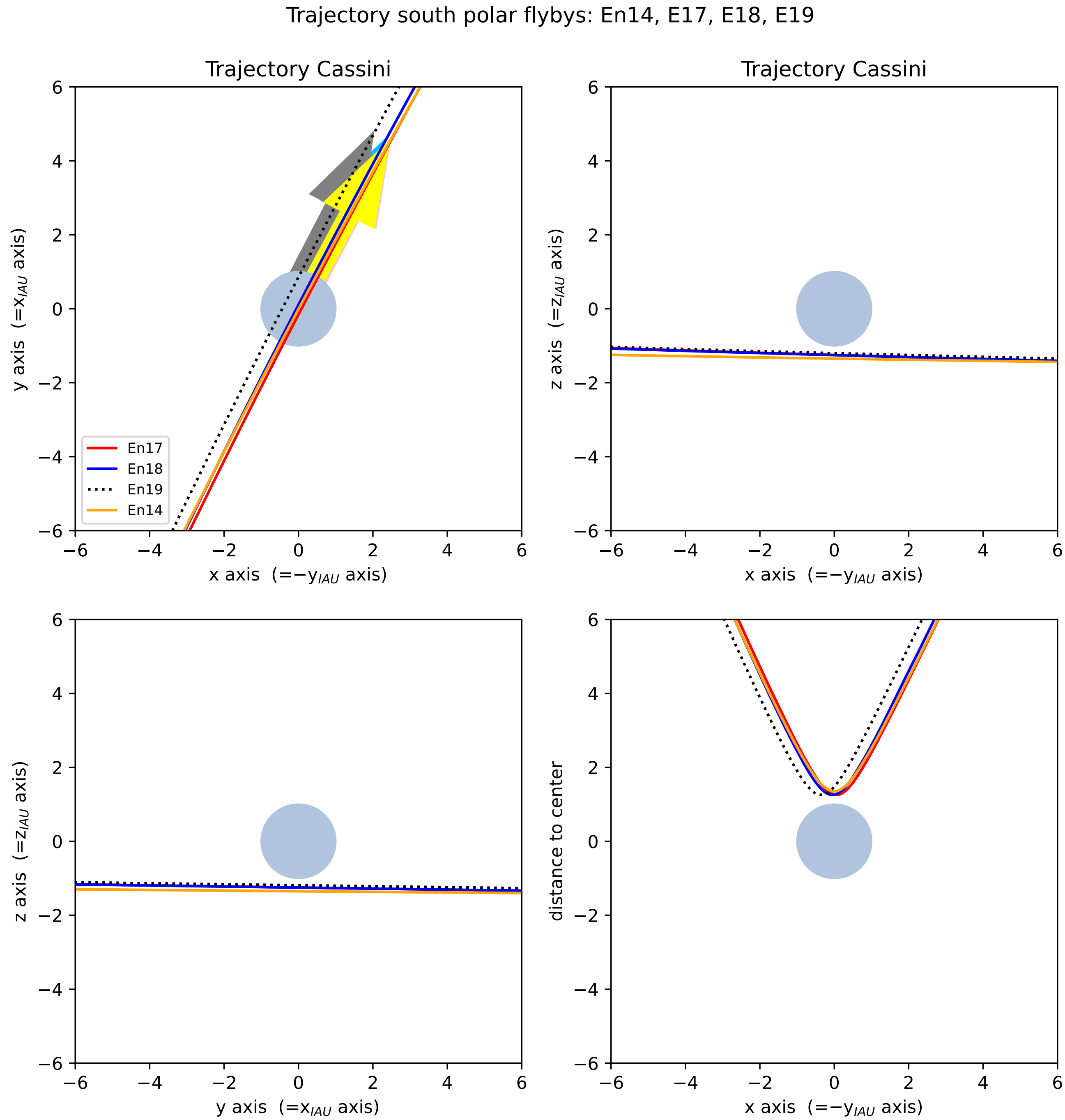

Figure 23. Trajectories of the close south polar flybys En14, En17, En18, and En19. The arrows indicate the direction of the Cassini spacecraft. Coordinates x, y, and z are in Enceladus IAU system rotated by 90° around the z-axis (definition see Section 5) with x approximately along Enceladus's orbital direction, y approximately towards Saturn, and z pointing north.

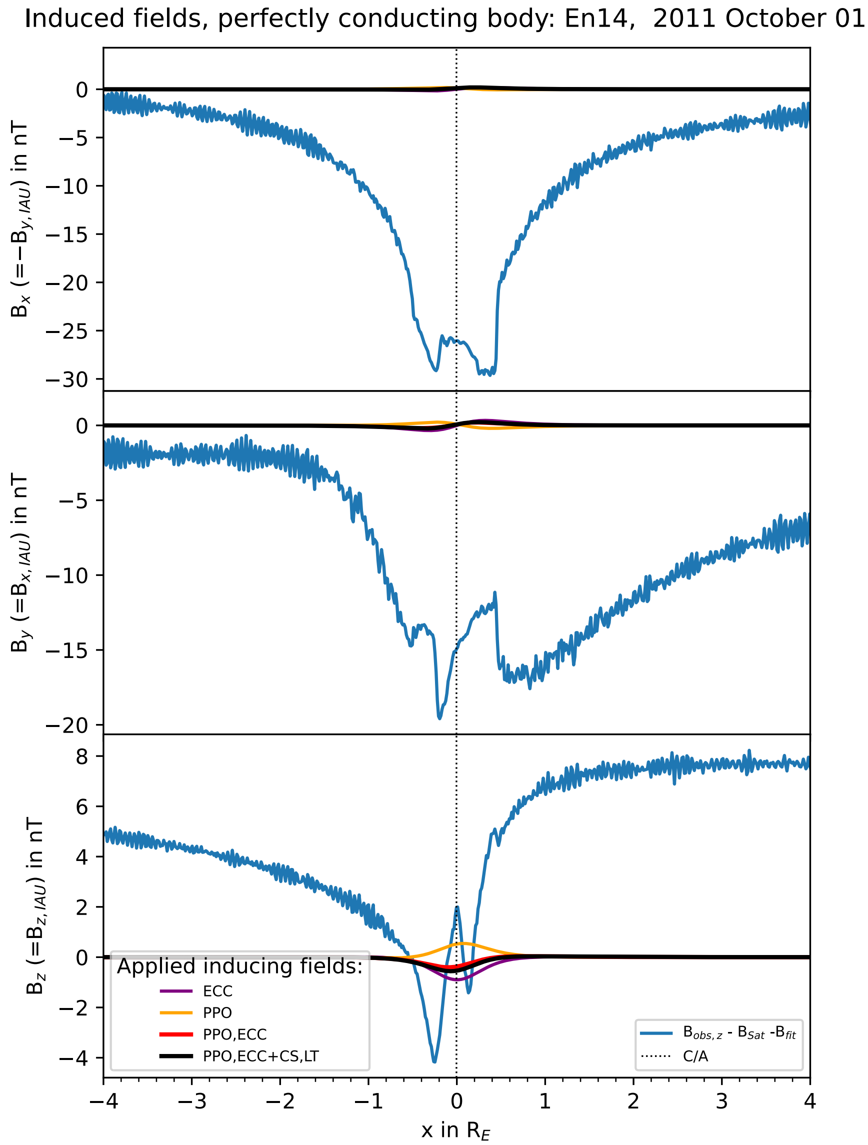

Figure 24. South polar flyby En14: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2.

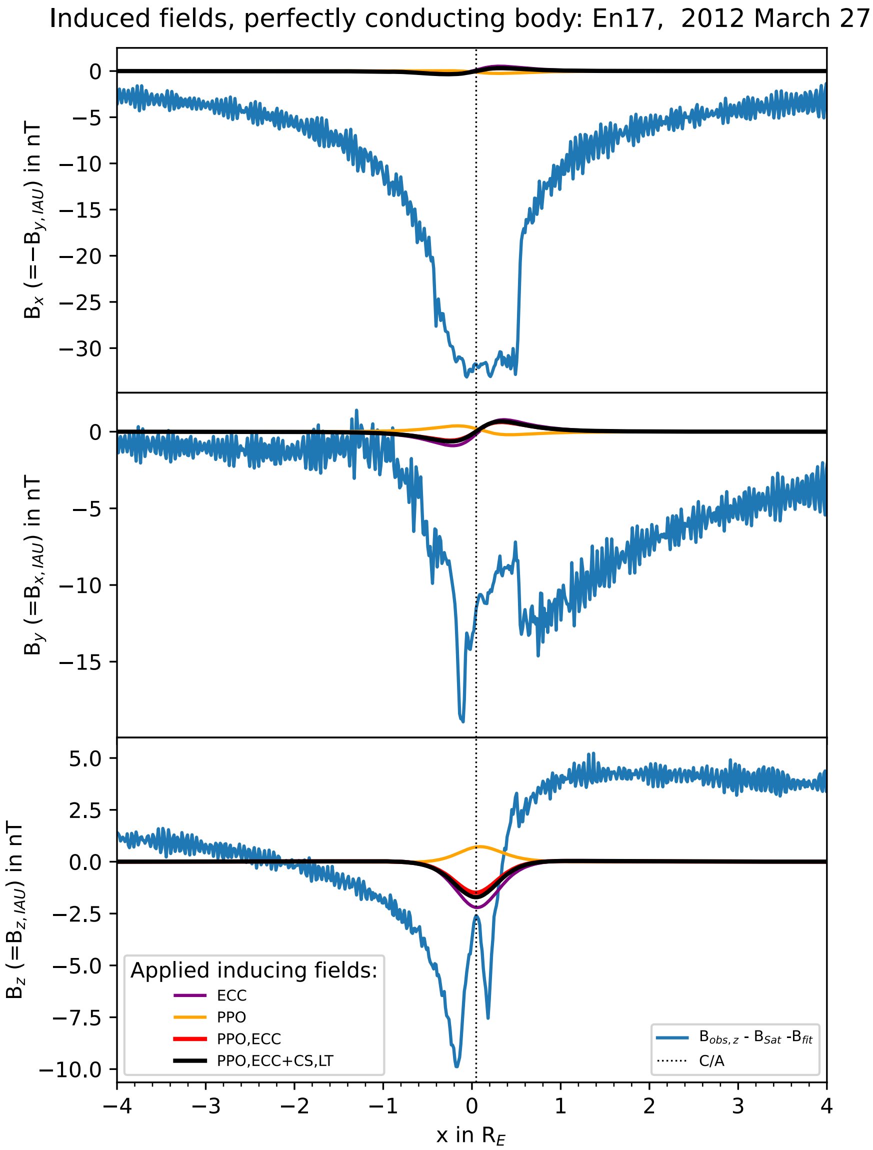

Figure 25. South polar flyby En17: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2.

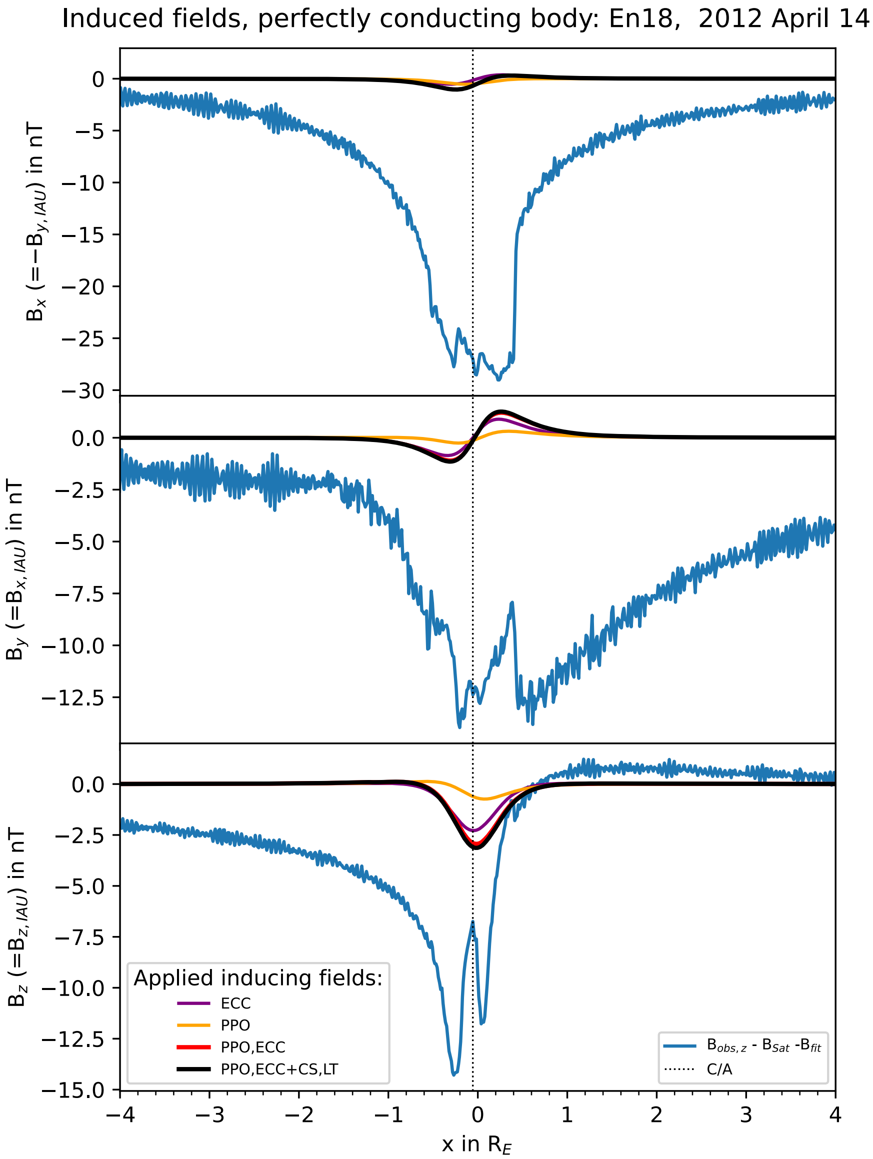

Figure 26. South polar flyby En18: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2.

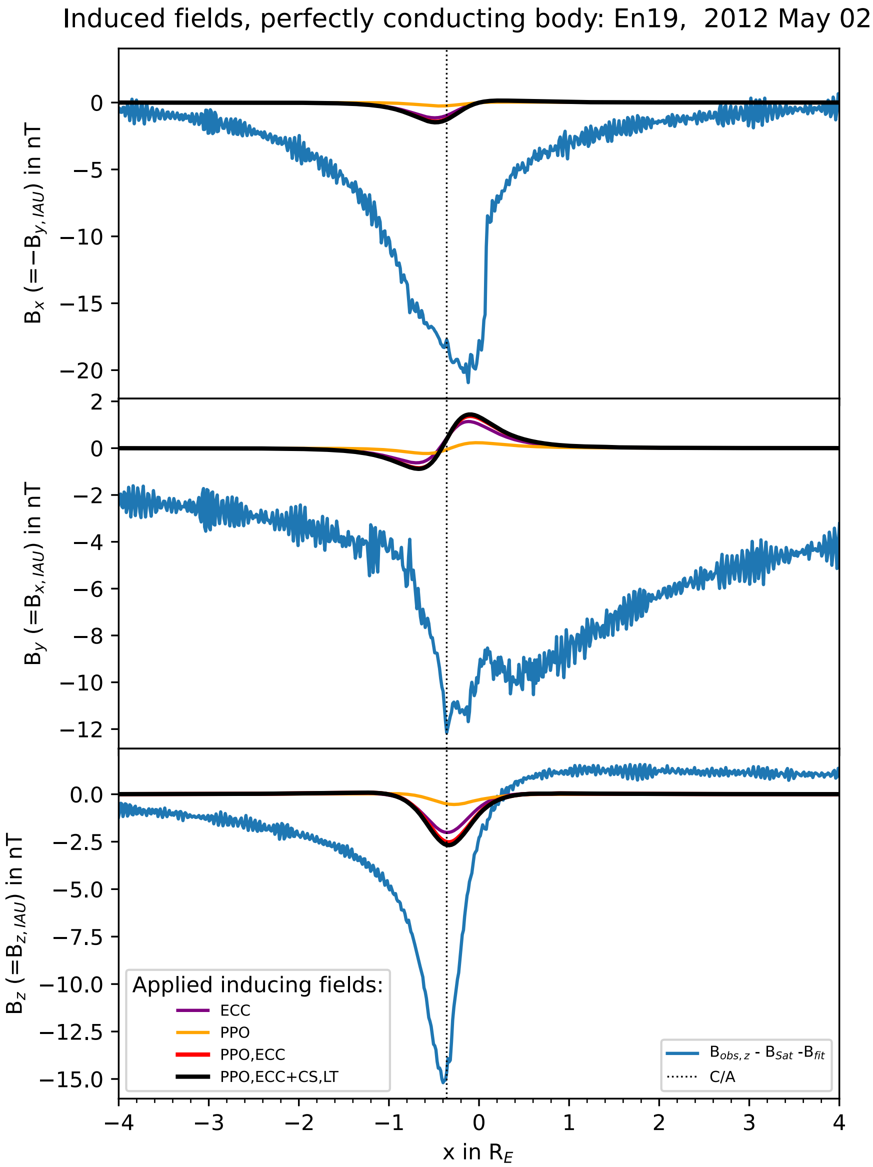

Figure 27. South polar flyby En19: magnetic field perturbations (blue) in rotated Enceladus IAU coordinates and standard IAU coordinates in parentheses. Colored lines referenced in the left legend display modeled induced fields for a perfectly conducting moon based on different models/components of the primary inducing field. The terminology is introduced in Section 3 and also used in Table 2.

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